

PADRE ISLAND NATIONAL SEASHORE

**REPLACE DAMAGED SECURITY BOLLARDS  
AT CLOSED BEACH**

**CORPUS CHRISTI, TEXAS  
PAIS 339194**

**PROJECT SPECIFICATIONS**



**NATIONAL PARK SERVICE  
INTER-MOUNTAIN REGIONAL OFFICE  
SERVING REGIONS 6, 7 & 8**

**MARCH 2026**

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## SECTION 01 11 00 - SUMMARY OF WORK

### PART 1 – GENERAL

#### 1.1 SUMMARY

A. Section includes the following:

1. Work Covered by Contract Documents
2. Contractor Use of Site
3. Public Use of Site
4. Conduct of Operations
5. Work Restrictions
6. Additional Reports

#### 1.2 WORK COVERED BY CONTRACT DOCUMENTS

A. Project Location: Padre Island National Seashore (PAIS). The work occurs at two sites along the Malaquite Beach (aka Closed Beach area). The bollards are located at the northern and southern limits of the seashore, approximately 4.65 miles apart (see Appendix A):

1. North Beach security access barriers (Lat: 272832.68N; Long: 971625.32W).
2. South Beach security access barriers (Lat: 272453.99N; Long: 97185.16W)

B. The Work consists of:

1. The replacement in-kind of twenty-three (23) damaged, missing and out of alignment fiberglass beach security bollards/pilings (12-inche diameter) to restore the integrity of the Closed Beach boundaries.
  - a. Tasks include replacing 23 failed/failing bollards entirely with 23 new bollards matching the existing in dimensions and material; manufactured fiberglass pilings resistant to weather, storm damage, and impact, including piling cone caps and fasteners.
  - b. The existing 15ft bollards were manufactured by Pearson Pilings, which are made from an epoxy/vinyl ester, with high elongation, low styrene monomer, high heat deflection temperature and chemical resistance. Any alternative piling/bollard types require prior approval from the Contracting Officer.
  - c. Installation will require that they are buried to no less than 8' in depth by mean of hydro jetting utilizing adjacent salt water sources to reduce impacts to surrounding natural resources at the two separate locations.
  - d. Replacement bollards will match existing with 12-inch diameter and black conical pylon caps.
2. The installation of 10 (ten) aluminum "Authorized Vehicles Only Beyond this Point" placards. At each bollard line, 5 (signs) will be attached onto the fiberglass bollards, utilizing approved and properly sized U-Bolts, instead of drilled in anchoring, so as not to deface the fiberglass bollards.

C. Period of Performance: Work shall be completed within 60 calendar days of the start date provided on the Notice to Proceed.

- a. Clarification: The Notice to Proceed will be issued with a period of performance, that is all inclusive of supply lead times, material deliveries, decommissioning, installation, anticipated average weather days for location, final commissioning, and acceptance of completed work. The contractor will be expected to provide schedule with adequate critical path details to ensure all work is completed within the stated calendar days from issuance of the Notice to Proceed. If weather impacts project beyond typical average weather patterns, contractor shall provide documentation in request for additional days.
- D. All work will be performed under a single contract.
- E. A site visit is highly encouraged to ensure verification of existing conditions and dimensions.
- 1.3 WORK PHASES (Not Used)
- 1.4 GOVERNMENT FURNISHED MATERIALS (Not Used)
- 1.5 CONTRACTOR USE OF SITE
  - A. General: Contractor shall have limited use of site for construction operations. Limit use of premises to areas within the construction limits indicated on drawings. Do not disturb portions of Project site beyond areas in which the Work is indicated.
    - 1. Confine construction operations to within 25 feet, either side of bollard line.
    - 2. Limit site disturbance to within 4 feet of bollards to be worked on.
  - B. Storage of Materials: Confine storage of materials to designated area approved by Park/CO.
  - C. Parking: Confine parking to the sections allowed for the public or as indicated by Park at each work site. The immediate area within 25 feet of bollard lines is authorized for contractor vehicle and equipment use, and for necessary for work operations/execution.
  - D. Stockpiling: Confine stockpiling to designated area approved by Park/CO.
  - E. Preservation of Natural Features and Protected Resources:
    - 1. Prevent damage to natural surroundings. Restore damaged areas, repair or replace damaged trees and plants, at no additional expense to the Government.
    - 2. Provide temporary barriers to protect existing trees and plants and root zones.
    - 3. Do not remove, injure, or harm any wildlife or marine life or other plants without prior approval. Park will ensure the removal of any animals that interfere with construction. There are several protected species that may be present at shoreline.
    - 4. Do not fasten ropes, cables, or guys to existing trees.
    - 5. Carefully supervise excavating, grading, filling, and other construction operations to prevent damage to nesting habitats etc.
  - F. Driveways and Entrances: Keep driveways and entrances serving premises clear and available to Government employees, and emergency vehicles at all times. Do not use for parking or storage of materials.

1. Schedule deliveries to minimize use of driveways and entrances.
  2. Schedule deliveries to minimize space and time requirements for storage of materials and equipment on-site.
- G. Construction Camp: Establishment of a camp within park will not be permitted.
- H. Hauling Restrictions: Comply with legal load restrictions in hauling of materials. Load restrictions on park roads are identical to state load restrictions with such additional regulations as may be imposed by the Park Superintendent. Information regarding rules and regulations for vehicular traffic on park roads may be obtained from the Office of the Park Superintendent. A special permit will not relieve Contractor of liability for damage which may result from moving of equipment.
- I. Bridge Restrictions: Identify jurisdictions, load restrictions, permit requirements, time and calendar restrictions as outlined.
- J. Pedestrian and/or Vehicular Traffic Control: Provide a traffic control plan for CO review, comment and acceptance if the project impedes, alters or affects the existing pedestrian and/or vehicular traffic. Products and procedures to be in accordance with Specification 015000, 3.3,C Traffic Controls.

#### 1.6 PUBLIC USE OF SITE

- A. The work area inside the park boundary will be closed to the public during construction, however there is public beach access on the outside of the boundary. Contractor shall provide a pedestrian and/or vehicular traffic control plan for CO review and acceptance if the project impedes, alters or affects the existing pedestrian and/or vehicular traffic. Contractor may utilize a proposed site plan or similar to indicate traffic control measures.
- B. Contractor shall conduct his operations to ensure the least inconvenience to public. Area closures may be permitted, when required, upon specific approval of the park, and coordination with the local authorities.

#### 1.7 CONDUCT OF OPERATIONS

- A. Contractor shall conduct his operations in conformance with rules and regulations promulgated by the Secretary of the Interior for the National Park Service, and applicable park rules and regulations prescribed by Park Superintendent.
- B. Work on Saturdays, Sundays, Federal holidays or at night may not be performed unless stated in the Work Restrictions below or without prior consent from the Contracting Officer. Submit requests 2 business/calendar days in advance of the work to the Contracting Officer for approval.
- C. No signs or advertisements (except those specified herein) shall be displayed on the construction site or within the park unless approved by the Contracting Officer.

#### 1.8 WORK RESTRICTIONS

- A. On-Site Work Hours: Work shall be generally performed during normal park business hours of 7 a.m. to 5 p.m., Monday through Friday, except when otherwise indicated.
- B. Existing Utilities: Not Used
- C. Extreme Fire Danger: If a Red Flag Warning or other extreme fire danger warning is issued that includes the project site, the Contractor shall coordinate with the Contracting Officer for any additional work restrictions that may be required.

#### 1.9 SPECIAL CONSTRUCTION REQUIREMENTS

- A. Include anticipated normal downtime related to weather as non-working time due to seasonal weather conditions in planning and scheduling work influenced by high and low ambient temperatures, wind, or precipitation to ensure completion of work within contract time. Anticipated weather days shall be based on data for local area from a reliable source like the National Oceanic and Atmospheric Administration (NOAA), National Park Service records, or source acceptable to Contracting Officer.
- B. Tide changes and adverse weather are likely to affect site conditions and contractor timelines for performing work.
- C. Carefully excavate around each existing bollard without any major impact to the beach or damage of the surrounding area.
- D. Remove the existing damaged fiberglass bollards that cannot cut or be sleeved with a large diameter bollard over the existing bollard.
- E. Transport the removed bollards to a designated disposal or recycling facility in compliance with environmental regulations.
- F. **Refer to attached compliance documents for additional information.**

#### 1.10 SOILS INVESTIGATION REPORT – Not Used

#### PART 2 – PRODUCTS (Not Used)

#### PART 3 – EXECUTION (Not Used)

#### **END OF SECTION 01 11 00**

### **SECTION 01 26 01 – CONTRACT MODIFICATION PROCEDURES**

#### PART 1 – GENERAL

#### 1.1 SUMMARY

- A. Section consists of administrative and procedural requirements for contract modifications.

## 1.2 DEFINITIONS AND ALLOWANCES

- A. Home Office Overhead: Costs incurred in support of all of a contractor's projects and not attributable to a specific job. The cost for home office overhead is only allowed as a percentage of all direct work excluding profit. The following items represent allowable home office overhead costs identified in Part 31 of the Federal Acquisition Regulation (FAR):

1. Rent
2. Utilities
3. Furnishings
4. Office equipment
5. Executive and management staff not exclusively assigned to the project
6. Support, accounting, and administrative staff
7. Preparation of cost proposals, estimating, and schedule analyses connected with Modifications
8. Estimating and preconstruction services
9. Mortgage costs
10. Real estate and corporate taxes
11. Automobile maintenance and travel costs for home office personnel
12. Home office insurances i.e., structure, automotive, umbrella, flood, etc.
13. Depreciation of equipment and other assets
14. Home office supplies (paper, staples, etc.)
15. Legal services
16. Accounting and data processing
17. Professional fees/registration

- B. General Conditions (Field Office Overhead): Management and administrative costs incurred on site for the designated project. Costs associated with preparation of modifications will not be allowed. Costs for these items are to be included only in the general conditions of the modification estimate. Only in the case of a contract time extension are additional general conditions included in modifications. The following items, if applicable, are considered allowable costs for calculating General Conditions:

1. Project Manager (PM), Assistant Project Manager
2. Superintendent, Assistant Superintendent
3. Quality Control, Safety Officer, Environmental Manager, etc.
4. Engineers
5. Travel, lodging, and per diem (as established by Federal Travel Regulations)
6. Scheduling
7. Field Office Trailers and associated temporary utilities
8. Field office supplies
  - a. Mailing and couriers
  - b. Reproduction costs
  - c. Storage
  - d. Phones
  - e. Computers
  - f. Copiers

9. Personal vehicles i.e., Superintendent Pickup trucks

- C. General Requirements: Costs directly associated with the project and are necessary to perform the actual work of the modification. These costs shall be shown as direct costs in the estimate. The following items, if applicable, are considered allowable costs for calculating General Requirements:

1. Hoisting
2. Material handling
3. Temporary fencing
4. Port-a-lets
5. Trash removal, dumpsters
6. Barricades
7. Small tools
8. Safety supplies
9. Scaffolding
10. Daily cleaning
11. Traffic control
12. Temporary signage
13. Temporary heating and power

- D. Personnel Costs: Costs included in the modification must only be for General Conditions staff and workers actually present and working on project site. Modification costs for salaried workers are only allowed within the structure of a 40-hour week and no overtime or holiday pay will be allowed.

1. Worker Hourly Rates are costs directly associated with the individual worker and consist of the following:
  - a. Base Rate: The hourly rate paid directly to the worker
  - b. Labor Burden: Employer payments of all applicable burdens; includes insurance and taxes the business must pay on behalf of the worker to government entities and educational forums, such as:
    - 1) Social Security
    - 2) Medicare
    - 3) Workers Compensation – Policy and company calculation to be made available.
    - 4) Federal Unemployment Tax Act (FUTA) - Cap Rate and percentage to be proportionally allocated over one year.
    - 5) State Unemployment Tax Act (SUTA) - Cap Rate and percentage to be proportionally allocated over one year.
    - 6) Union agreement costs - Other costs required under an enforceable collective bargaining agreement.
  - c. Fringe Benefits: Various non-wage compensations provided to employees such as:
    - 1) Health Care Insurance Premiums
    - 2) Cell Phone
    - 3) Clothing
    - 4) 401K and Pensions
    - 5) Vehicle allowances



- 6) Gas allowance
  - 7) Life insurance premiums
  - 8) Disability insurance
  - 9) Other Fringe Benefits required under an enforceable collective bargaining agreement
- E. Bonuses or Deferred Compensation: No Bonus or Deferred Compensation will be allowed within any components of pricing including Home Office Overhead, General Conditions, General Requirements, Hourly Worker Rates, or the direct costs of work.
  - F. General Liability Insurance: An insurance policy that protects Contractor from claims resulting from bodily injury or property damage to a third party. Include as a separate line item within all modification proposals and provide a current insurance quote upon request.
  - G. Performance and Payment Bonds: A performance bond is a surety bond issued by an insurance company or bank to guarantee satisfactory completion of a project. The Payment Bond guarantees the Contractor will pay the labor and material costs incurred. Banks and Insurance companies charge a premium for individual project based on a sliding scale related to the size of the project. Include as a separate line item in modification proposals and provide current company bonding rates upon request.
  - H. Builder's Risk Insurance: Covers the contractor's loss due to fire, high winds, or other natural forces. Not reimbursed by the National Park Service (NPS) and shall not be included in modification proposals.

### 1.3 MODIFICATION PROPOSAL PRICING REQUIREMENTS

#### A. General:

- 1. Proposal shall be received in the format and within the time frame specified in the Request for Proposal (RFP) letter. Costs or delays resulting from failure of contractor to submit within the time frame specified will not be compensable.
- 2. Proposal shall be detailed with itemized lists of equipment, materials, labor, production rates, overhead, profit, and bond markup for each item. Labor costs must be itemized by craft and hourly rate, including Fringe Benefits and Labor Burden. If the costs of Fringe Benefits and Labor Burden are not itemized, it is assumed they are included in the hourly rate shown, or contractor is not requesting reimbursement. Contractor may utilize the government provided Contractor Estimate Form, or their own form, provided that it contains the same information and level of detail as the Government's form.
- 3. Requests for extensions of contract time as a result of change must be justified with a Time Impact Analysis (TIA). Refer to Section 01 32 16 "Construction Schedule", for time impact analysis requirements. TIA and associated costs shall be received with the proposal by the date shown within the Request for Proposal letter. Contractor's failure to submit within the specified time frame will be construed as the Contractor waiving right for additional time and no time extension will be allowed.
- 4. All supporting documentation used to justify the proposed modification will be made available to the Contracting Officer (CO) upon request.
- 5. Contractor shall review and approve all subcontractor/supplier pricing in detail for proper format, scope, production rates, and pricing prior to submission to

NPS. All delay costs associated with not reviewing and approving subcontractor/supplier pricing will be borne by the Contractor.

6. All pricing and production rates within the estimate must be based on fair and reasonable pricing and cannot include built-in contingency.

B. Labor:

1. Contractor shall estimate cost of labor by itemizing each craft involved, indicating worker hourly rate (base rate + labor burden + fringe benefits) for each and itemizing hours required for each craft directly engaged in modification work. Any work proposed requiring overtime work or premium pay shall be itemized separately. Rates shall be in accordance with the Davis-Bacon Act as incorporated herein. Labor Burden may include payroll taxes, Social Security, unemployment insurances, workers compensation insurance, Federal Insurance Contributions Act (FICA), FUTA, and other direct costs resulting from Federal, State or local laws.
2. Itemize labor costs for equipment operators separate from equipment costs.
3. Labor cost for foremen shall only be costs for related work required for the modification.

C. Materials:

1. Estimated cost for materials shall include quotes from multiple sources. Material prices shall include applicable fees and credits, including but not limited to, sales tax, freight and delivery charges, and tax rebates.
2. No markup shall be applied to any material provided by NPS.

D. Equipment:

1. Equipment used for the project must be appropriately sized for work being performed.
2. Do not include costs for "miscellaneous tools and equipment", in your proposal for a replacement value of \$500 or less. Costs shown in excess of \$500 shall be broken out separately.
3. Regardless of ownership, rates to be used in determining equipment rental costs shall be the lowest cost from one of the following sources:
  - a. United States (U.S.) Army Corps of Engineers, Ownership and Operating Expense Schedule (use latest edition and applicable region)
  - b. Construction Blue Book
  - c. Local equipment rental rates, documented by actual invoice charges, or itemized vendor quotes.
4. Estimated equipment rates shall include operating costs of all fuel, oil, lubrication, supplies, small tools, necessary attachments, ground engaging components, tires and tracks, routine repairs and maintenance (cost of major repair and overhaul for rental equipment is not allowed per Federal Acquisition Regulation (FAR) 31.105(d)(2)), depreciation, storage, insurance, and all incidentals. Mobilization, if applicable, may be included for equipment solely used on the modification work but must be listed separately.
5. Estimate full rate for equipment only for duration that equipment will be utilized to accomplish work of the modification.
6. Standby unit rates used in accordance with paragraph 1.3, D, 2, above. If the

U.S. Army Corp of Engineers is utilized, then their standby rates prevail. If Bluebook or local equipment pricing is accepted, then 1/2 of equipment costs minus any operating costs, major repair and overhaul will be accepted.

7. If equipment is in standby mode due solely to a documented NPS delay, established standby rate shall apply from the first day of the delay.
8. Equipment not used and on job site for up to five consecutive days may be classified at standby rates, provided the equipment is or has been used solely to perform work on the modification and will be necessary to complete additional modification work. Equipment still on the jobsite but not in use after five consecutive days will not be considered in the modification pricing.
9. Requests for compensation for equipment stand by time must be justified, documented and itemized separately.
10. The estimated timeframe (daily, weekly, monthly) for use of the equipment must reflect the lowest cost to the Government.

## PART 2 – PRODUCTS

## PART 3 – EXECUTION

### **END OF SECTION 012601**

## **SECTION 01 27 00 – DEFINITION OF CONTRACT LINE ITEMS**

### PART 1 – GENERAL

#### 1.1 SUMMARY

- A. Section explains in general, what is and is not included in a contract line item, and limits or cut-off points where one item ends, and another begins.
- B. If no contract line item exists for a portion of work, include costs in a related item.

### PART 2 – PRODUCTS

### PART 3 – EXECUTION

#### 3.1 LIST OF CONTRACT LINE ITEMS

- A. Contract Line-Item Number 01: Selective Demolition of 23 Existing Piles
  1. This item consists of all associated costs for the proper removal of 23 damaged fiberglass bollards, and the legal disposal of construction waste & any hazardous materials.
  2. Payment will be made at the contract lump sum price.

- B. Contract Line-Item Number 02: Install 23 New Fiberglass Piles
  - 1. This item consists of all associated costs for the proper installation of 23 new fiberglass bollards, (Pearson Pilings 10” to 12” or approved equal) and the legal disposal of construction waste & any hazardous materials.
  - 2. Payment will be made at the contract lump sum price.
- C. Contract Line-Item Number 03: Install 10 New Signs
  - 1. This item consists of all associated costs for the purchase and installation of 10 new aluminum "Authorized Vehicles Only Beyond this Point" placards and the legal disposal of construction waste & any hazardous materials.
    - a. At each bollard line (north & south), 5 placards will be clamped onto pre-selected fiberglass bollards, instead of drilled in anchoring, so as not to deface the fiberglass bollard.
  - 2. Payment will be made at the contract lump sum price.
- D. Contract Line-Item Number 04 Mobilization
  - 1. This item consists of all associated costs for contractor mobilization to both work sites, the proper set up and tear down of equipment, contractor demobilization and the legal disposal of construction waste & any hazardous materials incidental to mobilization operations.
  - 2. Payment will be made at the contract lump sum price.

**END OF SECTION 01 27 00**

**SECTION 01 31 00 - PROJECT MANAGEMENT AND COORDINATION**

**PART 1 – GENERAL**

**1.1 SUMMARY**

- A. Section includes administrative provisions for coordinating construction operations on Project including, but not limited to, the following:
  - 1. Definitions
  - 2. Construction Coordination
  - 3. Submittals
  - 4. Coordination Drawings
  - 5. Requests for Information (RFIs)
  - 6. NPS Project Website
  - 7. Project Meetings
  - 8. Environmental Coordination
  - 9. Permits
- B. Related Requirements:

1. Section 01 32 16 "Construction Schedule" for preparing and submitting Contractor's construction schedule.
2. Section 01 73 40 "Execution" for procedures for coordinating general installation and field-engineering services, including establishment of benchmarks and control points.
3. Section 01 77 00 "Closeout Procedures" for coordinating closeout of the Contract.

## 1.2 DEFINITIONS

- A. Agency with Jurisdiction
- B. Construction Permits – Contractor Provided
- C. Government Furnished Permits

## 1.3 CONSTRUCTION COORDINATION

- A. Coordination: Coordinate construction operations included in different Sections of Specifications to ensure efficient and orderly installation of each part of the Work. Coordinate construction operations, included in different Sections, which depend on each other for proper installation, connection, and operation.
  1. Schedule construction operations in sequence required to obtain best results where installation of one part of the Work depends on installation of other components, before or after its own installation.
  2. Properly plan construction operations to include permit requirements. Allow enough time to execute permit provisions to maintain work schedule, site visits, inspections, and reporting deadlines.
- B. Administrative Procedures: Coordinate scheduling and timing of required administrative procedures with other construction activities and activities of other contractors to avoid conflicts and to ensure orderly progress of the Work. Such administrative activities include, but are not limited to:
  1. Preparation of Contractor's Construction Schedule
  2. Preparation of the Schedule of Values
  3. Installation and removal of temporary facilities and controls
  4. Delivery and processing of submittals
  5. Progress meetings
  6. Permit requirements
  7. Pre-installation conferences
  8. Project closeout activities
  9. Commissioning activities

## 1.4 SUBMITTALS

- A. Division 1 documents: The following items shall be submitted a minimum of one week prior to-Preconstruction Conference. Contracting Officer will notify Contractor of tentative date for Pre-Construction Conference.

1. Letter designating Project Superintendent
  2. Construction Schedule
  3. A Comprehensive Schedule of Values
  4. Accident Prevention Plan
  5. A List of Subcontractors for this project
  6. SF1413 for Contactor and all Subcontractors Written statements from Subcontractors certifying compliance with applicable labor standard clauses.
  7. Certificates of Insurance for Contactor and all Subcontractors
  8. Quality Control Plan
  9. Pedestrian/Vehicular Traffic Control Plan
  10. List of Required Construction Permits. Include the following information for each permit:
    - a. Name of Permit
    - b. Agency(ies) with Jurisdiction issuing the permit
    - c. Information required from Government to complete permit application
- B. Provide items listed to Contracting Officer before Pre-Construction Conference. If all documents have not been received one week prior to scheduled Pre-Construction Conference date, conference may be cancelled, Notice to Proceed may not be issued, and Contracting Officer will consider other contractual remedies. Work shall not commence until written Notice to Proceed has been issued.

#### 1.5 REQUESTS FOR INFORMATION (RFIs)

- A. General: Immediately on discovery of the need for additional information or interpretation of Contract Documents, Contractor shall prepare and submit an RFI with content of Paragraph 1.5.B below.
1. Contracting Officer will not respond to RFIs submitted by other entities controlled by Contractor.
  2. Coordinate and submit RFIs in a prompt manner to avoid delays in the work.
- B. Content of RFI: Include detailed, legible description of item needing information or interpretation and the following:
1. RFI number, numbered sequentially
  2. Date
  3. RFI subject
  4. Specification Section number and title and related paragraphs, as appropriate.
  5. Drawing number and detail references, as appropriate.
  6. Field dimensions and conditions, as appropriate.
  7. Contractor's suggested resolution: If Contractor's suggested resolution impacts Contract Time or Contract Sum, Contractor shall state impact in RFI.
  8. Contractor's signature
  9. Requested date for response
  10. Attachments: Include sketches, descriptions, measurements, photos, Product Data, Shop Drawings, coordination drawings, and other information necessary to fully describe items needing interpretation.

- a. Include dimensions, thicknesses, structural grid references, and details of affected materials, assemblies, and attachments on attached sketches.
- C. Contracting Officer's Action: Contracting Officer will review each RFI, determine action required, and respond. Contracting Officer will determine critical nature of each RFI and issue response accordingly.
  - 1. The following are not considered to be RFIs and will receive no action:
    - a. Requests for approval of submittals.
    - b. Requests for approval of substitutions.
    - c. Requests for approval of Contractor's means and methods.
    - d. Requests for coordination information already indicated in Contract Documents.
    - e. Requests for adjustments in Contract Time or Contract Sum.
    - f. Requests for interpretation of Architect/Engineer's actions on submittals.
    - g. Incomplete RFIs or inaccurately prepared RFIs.
  - 2. Contracting Officer's action may include a request for additional information; time for response will date from time of receipt of additional information.
  - 3. Contracting Officer's action on RFIs may result in need for a change to Contract Time or Contract Sum. All contract changes will be processed following terms and conditions of contract.

#### 1.6 PROJECT MEETINGS

- A. Preconstruction Conference: Before start of construction, Contracting Officer will arrange an on-site meeting with Contractor. Meeting agenda will include the following as a minimum:
  - 1. Roles & Responsibilities / Lines of Authority
  - 2. Park rules and regulations
  - 3. Jobsite Safety
  - 4. Resolution of comments on required Division 1 documents
  - 5. Coordination of Subcontractors
  - 6. Labor law application
  - 7. Modifications
  - 8. Payments to Contractor
  - 9. Payroll reports
  - 10. Contract time
  - 11. Liquidated damages
  - 12. Contractor Performance Evaluation
  - 13. Display of Hotline posters
  - 14. Notice to proceed
  - 15. Correspondence procedures
  - 16. Acceptance/rejection of work
  - 17. Progress meetings
  - 18. Submittal procedures
  - 19. NPS Final Accessibility Inspection
  - 20. Environmental requirements

21. Permit requirements
22. As-constructed drawings/operation and maintenance (O&M) manuals.
23. Saturday, Sunday, holiday and night work.
24. Reference materials
25. Value engineering
26. Schedule of Values

B. Progress Meetings: Contracting Officer will schedule weekly meetings with Contractor, as required.

1. Attendees: In addition to Government Representatives, the Contractor and any , Subcontractor, supplier, and other entity deemed necessary for attendance shall be represented. Participants at meeting shall be familiar with Project and authorized to conclude matters relating to the Work.
2. Agenda: Review and correct or approve minutes of previous progress meeting. Review other items of significance that could affect progress. Meeting agenda will include:
  - a. Approval of minutes of previous meetings
  - b. Submittal status
  - c. Review of off-site fabrication and delivery schedules.
  - d. Requests for information (RFI) and other issues.
  - e. Modifications
  - f. Work in progress and projected; review 3-week look ahead schedule.
    - 1) Status of required inspections (Special Inspections, Accessibility, etc.)
  - g. Inspections of work in progress and projected (Special inspections, Accessibility, etc.)
  - h. Construction Schedule update (provide updated Critical Path Method (CPM)).
  - i. Status of Project Record Drawings and O&M manuals.
  - j. Other business relating to work.
  - k. Permit requirements
3. Facilitator: A designated meeting facilitator is to prepare and distribute the agendas in advance of the meetings, facilitate and take notes during the meetings, prepare and distribute the minutes for review subsequent to the meetings, finalize meeting minutes and ensure the 3week look ahead schedule and other attachments as necessary are distributed in advance of the meetings.

C. Preinstallation Conferences: Conduct at Project site before each construction activity that requires coordination with other construction.

1. Attendees: Contractor, Installer and representatives of manufacturers and fabricators involved. Advise Contracting Officer of scheduled meeting dates.
2. Agenda: Review progress of other construction activities and preparations for particular activity under consideration, including requirements for:
  - a. Contract Documents
  - b. Options
  - c. Related RFIs
  - d. Related Change Orders



- e. Purchases
  - f. Deliveries
  - g. Submittals
  - h. Possible conflicts
  - i. Compatibility requirements
  - j. Time schedules
  - k. Weather limitations
  - l. Manufacturer's written instructions
  - m. Warranty requirements
  - n. Compatibility of materials
  - o. Acceptability of substrates
  - p. Temporary facilities and controls
  - q. Space and access limitations
  - r. Regulations of agency(ies) with jurisdiction
  - s. Testing and inspecting requirements
  - t. Installation procedures
  - u. Coordination with other work
  - v. Required performance results
  - w. Protection of adjacent work
  - x. Protection of construction and personnel
- 3. Record significant conference discussions, agreements, and disagreements, including required corrective measures and actions.
  - 4. Reporting: Distribute minutes of meeting to each party present and to other parties requiring information.
  - 5. Do not proceed with installation if conference cannot be successfully concluded. Initiate whatever actions are necessary to resolve impediments to performance of Work and reconvene conference at earliest feasible date.

## 1.7 PERMITS

- A. Government Furnished Permits: During development of the project's design, permits listed below were negotiated and agreed to by the Government. Terms and provisions of these permits shall be adhered to for the duration specified in each permit.
  - 1. US Army Corps of Engineers (USACE) Nationwide Maintenance permit (NWP) for limited in kind replacement of individual damaged bollards. Maintenance:
    - a. The repair, rehabilitation, or replacement of any previously authorized, currently serviceable structure or fill, or of any currently serviceable structure or fill authorized by 33 CFR 330.3, provided that the structure or fill is not to be put to uses differing from those uses specified or contemplated for it in the original permit or the most recently authorized modification. Minor deviations in the structure's configuration or filled area, including those due to changes in materials, construction techniques, requirements of other regulatory agencies, or current construction codes or safety standards that are necessary to make the repair, rehabilitation, or replacement are authorized. This NWP also authorizes the removal of previously authorized structures or fills. Any stream channel modification is limited to the minimum necessary for the

repair, rehabilitation, or replacement of the structure or fill; such modifications, including the removal of material from the stream channel, must be immediately adjacent to the project. This NWP also authorizes the removal of accumulated sediment and debris within, and in the immediate vicinity of, the structure or fill. This NWP also authorizes the repair, rehabilitation, or replacement of those structures or fills destroyed or damaged by storms, floods, fire or other discrete events, provided the repair, rehabilitation, or replacement is commenced, or is under contract to commence, within two years of the date of their destruction or damage. In cases of catastrophic events, such as hurricanes or tornadoes, this two-year limit may be waived by the district engineer, provided the permittee can demonstrate funding, contract, or other similar delays.

- b. Not Used
- c. This NWP also authorizes temporary structures, fills, and work, including the use of temporary mats, necessary to conduct the maintenance activity. Appropriate measures must be taken to maintain normal downstream flows and minimize flooding to the maximum extent practicable, when temporary structures, work, and discharges of dredged or fill material, including cofferdams, are necessary for construction activities, access fills, or dewatering of construction sites. Temporary fills must consist of materials, and be placed in a manner, that will not be eroded by expected high flows. After conducting the maintenance activity, temporary fills must be removed in their entirety and the affected areas returned to pre-construction elevations. The areas affected by temporary fills must be revegetated, as appropriate.
- d. For activities authorized by paragraph (b) of this NWP, the permittee must submit a pre-construction notification to the district engineer prior to commencing the activity (see general condition 32).

## PART 2 – PRODUCTS

## PART 3 – EXECUTION

### **END OF SECTION 01 31 00**

## **SECTION 01 32 16 – CONSTRUCTION SCHEDULE**

### PART 1 – GENERAL

#### 1.1 SUMMARY

- A. Section consists of Construction Schedule requirements including:

1. Schedule of Values
  2. Construction Schedule Requirements.
  3. Construction Schedule Updates.
  4. Time Impact Analysis.
- B. Purpose: The Construction Schedule ensures adequate planning, coordination, scheduling, and reporting during execution of the work by the Contractor. It shall assist the Contractor and Contracting Officer (CO) in monitoring the progress of the work, evaluating proposed changes, and processing Contractor's monthly progress payments. It shall include the dates in the contract, phases, milestones, occupancies, holidays, weather consideration, a critical path, and the requirements of this section.

## 1.2 DEFINITIONS

- A. Activity: A discrete part of a project identified for planning, scheduling, monitoring, and controlling the construction project. Activities included in a construction schedule consume time and resources.
1. Critical activities are activities on the critical path. They start and finish on the planned early start and finish times.
  2. Predecessor Activity: An activity that precedes another activity in the network.
  3. Successor Activity: An activity that follows another activity in the network.
- B. Cost Loading: Allocation of the Schedule of Values for completion of an activity as scheduled. The sum of costs for all activities must equal the total Contract Sum, unless otherwise approved by the Contracting Officer.
- C. Critical Path Method (CPM): Method of planning and scheduling a construction project where activities are arranged based on activity relationships. Network calculations determine when activities can be performed and the critical path of Project.
- D. Critical Path: Longest connected chain of interdependent activities through the network schedule that establishes minimum overall Project duration and contains no float.
- E. Float: Measure of leeway in starting and completing an activity.
1. Float: Not for the exclusive use or benefit of the Government or Contractor but is jointly owned.
  2. Free Float: Amount of time an activity can be delayed without adversely affecting the early start of the successor activity.
  3. Total Float: Measure of leeway in starting or completing an activity without adversely affecting planned Project completion date.
- F. Resource Loading: Allocation of manpower and equipment necessary for completion of an activity as scheduled.
- G. Fragnet: A partial or fragmentary network that breaks down activities into smaller activities for greater detail.

### 1.3 SUBMITTALS

- A. Electronic Copies: Schedules and reports submitted shall be in native electronic file formats and/or in PDF format. The intent of the Government is to limit the number of printed reports to those determined by the project team as essential.
- B. Schedule of Values: After contract award and before Pre-Construction conference, submit schedule of dollar values based on Contract Price Schedule.
- C. Construction Baseline Schedule: After contract award and before Pre-Construction conference, submit the baseline schedule in electronic format (native file and PDF) formatted to show the entire schedule for the entire construction period. Utilize Schedule of Values in preparation of Construction Baseline Schedule.
- D. Critical Path Method (CPM) Reports: Concurrent with CPM schedule, submit in electronic format (PDF) the following computer-generated reports. For each activity, include activity number, activity description, cost and resource loading, original duration, remaining duration, early start date, early finish date, late start date, late finish date, and total float in calendar days.
  - 1. Activity Report: List of activities sorted by activity number and then early start date, or actual start date if known.
  - 2. Logic Report: List of predecessor and successor tasks for activities sorted in ascending order by activity number and then early start date, or actual start date if known.
  - 3. Total Float Report: List of activities sorted in ascending order of total float.
- E. Construction Schedule Updates: Provide as requested by the Contracting Officer.
- F. Construction Schedule Revisions and Time Impact Analysis: Provide as requested by the Contracting Officer. Incorporate a Fragmentary Network (Fragnet) into currently accepted Construction Schedule that demonstrates how Contractor proposes to incorporate a modification, change, delay, or Contractor request.

### 1.4 QUALITY ASSURANCE

- A. Contractor shall meet with Contracting Officer on day of the preconstruction conference to go over:
  - 1. Discuss constraints, including work stages, area separations, interim milestones and substantial completions.
  - 2. Review time required for review of submittals and re-submittals.
  - 3. Review requirements for tests and inspections by independent testing and inspecting agencies.
  - 4. Review time required for completion and startup procedures.
  - 5. Review time required for obtaining and activating permits.
  - 6. Review and finalize list of construction activities to establish the baseline schedule.
  - 7. Discuss reporting requirements and protocol for naming and transmitting electronic submittals.

## 1.5 COORDINATION

- A. Coordinate preparation and processing of schedules and reports with performance of construction activities and with scheduling and reporting of separate Contractors.
- B. Coordinate Construction Baseline Schedule with Schedule of Values, list of subcontracts, Submittals Schedule, progress reports, payment requests, and other required schedules and reports.
  - 1. In developing Construction Baseline Schedule, ensure Subcontractor's work at all tiers, and prime Contractor's work, is included and coordinated.
  - 2. Secure time commitments for performing critical elements of work from parties involved.
  - 3. Coordinate each construction activity in network with other activities and schedule in proper sequence.

## PART 2 – PRODUCTS

### 2.1 SCHEDULE OF VALUES

- A. Breakdown each lump-sum item into component work activities used in the schedule for which progress payments may be requested. Work activities broken out within schedule of values may be integrated into and made a logical part of the construction baseline schedule. Total costs for the component work activities shall equal contract price for that lump-sum item. Contracting Officer may request data to verify accuracy of dollar values. Include mobilization, general condition costs, overhead and profit in the total dollar value of unit price items and in the component work activities for each lump-sum item. Do not include mobilization, general condition costs, overhead or profit as a separate item.
- B. Do not break down unit price items. Use only the contract price for unit price items.
- C. Total cost of all items shall equal the contract price. The Schedule of Values will form the basis for progress payments and the Construction Schedule.

### 2.2 CONSTRUCTION SCHEDULE REQUIREMENTS

- A. Construction Baseline Schedule: Prepare Construction Baseline Schedule using a computerized, cost and resource-based, time-scaled Critical Path Method network analysis diagram for the Work.
  - 1. Develop and finalize Construction Baseline Schedule so it can be accepted for use no later than 30 days after date established for the Notice of Award.
    - a. Failure to include any work item required for performance of this Contract shall not excuse Contractor from completing work within applicable completion dates, regardless of Government's acceptance of schedule.

2. Establish procedures for monitoring and updating Construction Baseline Schedule and for reporting progress. Coordinate procedures with progress meeting and payment request dates.
- B. Construction Baseline Schedule Preparation: Prepare a list of all activities required to complete the Work. Using preliminary Critical Path Method network diagram, prepare a skeleton network to identify probable critical paths.
1. Activities: Indicate estimated duration, sequence requirements, and relationship of each activity in relation to other activities.
  2. Critical Path Activities: Identify critical path activities, including those for interim completion dates. Scheduled start and completion dates shall be consistent with Contract milestone dates.
  3. Processing: Process data to produce output data on a computer-drawn, time-scaled network. Revise data, reorganize activity sequences, and reproduce as often as necessary to produce the Critical Path Method schedule within the limitations of the Contract Time.
  4. Show sequence and interdependence of activities required for completion of work. Ensure work sequences are logical, and Construction Baseline Schedule shows a coordinated plan of the work.
  5. Include anticipated normal downtime related to weather as non-working time due to seasonal weather conditions in planning and scheduling work influenced by high and low ambient temperatures, wind, or precipitation to ensure completion of work within contract time. Anticipated weather days shall be based on data for local area from a reliable source like the National Oceanic and Atmospheric Administration (NOAA), National Park Service records, or source acceptable to Contracting Officer.
  6. Time Frame: Proposed duration assigned to each activity shall be Contractor's best estimate of time required to complete activity considering the scope and resources planned for activity.
    - a. An early finish date may be shown but the late finish date shall be same date as last day of contract period. An early completion schedule shall contain:
      - 1) Insert an activity titled "Project Float" as a successor to last activity in early project completion schedule network.
      - 2) Add a milestone titled "Contract End Date" as a successor to the activity "Project Float".
      - 3) Add duration to the activity "Project Float" as required so the milestone "Contract End Date" equals the last day of Contract Period.
    - b. Contract completion date shall not be changed by submission of a schedule that shows an early completion date.
    - c. Contractor shall limit use of lead or lag durations between schedule activities.
    - d. Activity Duration: Define so no activity is longer than 14 days, except for non-construction activities including mobilization, shop drawings and submittals, fabrication and delivery of materials and equipment.
    - e. Procurement Activities: Include procurement process activities for long lead items and major items, requiring a cycle of more than 60 calendar days, as

separate activities in the schedule. Procurement cycle activities can include submittals, approvals, purchasing, fabrication, and delivery.

- f. Submittal Review Time: Include review and re-submittal times indicated. Coordinate submittal review times in Construction Baseline Schedule.
- g. Substantial Completion: Allow time for Government administrative procedures necessary for certification of Substantial Completion. (For more information, refer to Specification 01 77 00 "Closeout Procedures.")

C. Joint Review, Revision, and Acceptance:

- 1. Within seven calendar days of receiving Contractor's proposed Construction Baseline Schedule, Contracting Officer shall review initial Construction Baseline Schedule.
- 2. Within seven calendar days after review, Contractor shall revise and resubmit Construction Baseline Schedule in accordance with comments presented from review.
- 3. In the event the Contractor fails to define any element of work, activity, or logic, and the Contracting Officer review does not detect this omission or error, such omission or error, when discovered by Contractor or Contracting Officer, shall be corrected by Contractor within seven calendar days and shall not affect contract period.
- 4. Upon Contracting Officer's acceptance of the Construction Baseline Schedule, save the schedule as a baseline and update on a monthly basis. Construction schedule updates will be used to evaluate Contractor's monthly applications for payment based upon information developed at monthly Construction Schedule update meetings.

D. Recovery Schedule: When periodic schedule update indicates Work is 10 or more calendar days behind current accepted schedule, a separate recovery schedule indicating means by which Contractor intends to regain compliance with the schedule shall also be submitted. Indicate changes to working hours, working days, crew sizes, and equipment required to achieve compliance, and date by which recovery shall be accomplished.

E. Computer Software: Prepare schedules using a program developed specifically to manage construction schedules.

- 1. Use Microsoft Project or Primavera.

## PART 3 – EXECUTION

### 3.1 CONSTRUCTION SCHEDULE UPDATES

- A. Progress Meeting Updates: Provide a three (3) week look-ahead schedule, derived from the currently accepted schedule, before each weekly progress meeting. Utilize look-ahead schedule to facilitate and take notes on discussions held during progress meeting.
- B. Monthly Schedule Updates:

1. General: Update the Construction Schedule on a monthly basis to reflect actual construction progress and activities throughout the entire contract period and until the project reaches the substantial completion milestone. The status date of each schedule update shall be the 7th day preceding the progress payment request date.
  2. Procedure: Contractor shall meet with the Contracting Officer each month at either a Construction Schedule update meeting or during a weekly Progress Meeting to review progress made to date and discuss any changes to the baseline schedule.
    - a. Review actual progress made during update period; actual and potential delaying activities; impediments to progress; issues related to inclement weather; progress toward established milestones and project float. Discussion shall include a review of work anticipated to be performed in the next month. Revisions to the schedule are to be identified for evaluation and acceptance or rejection.
  3. As Work progresses, indicate Actual Completion percentage for each activity.
  4. Progress Payments: Monthly updating of the currently accepted Construction Schedule shall be an integral part of the process upon which progress payments will be made. If Contractor fails to provide schedule updates or revisions, a portion of the monthly payment may be retained until corrections have been made.
- C. Distribution: Distribute copies of accepted schedule to the Contracting Officer, Contracting Officers Representative, Subcontractors, testing and inspecting agencies, and other parties identified by Contractor with a need-to-know schedule responsibility. Also post a copy in the Contractor's field office.
1. When revisions are made, distribute updated schedules to same parties and post in same locations. Delete parties from distribution when they have completed their assigned portion of the Work.
- D. Construction Schedule Revisions:
1. Required Revisions: If, as a result of the monthly schedule update, it appears the currently accepted Construction Schedule no longer represents actual prosecution and progress of the work, Contracting Officer will request, and Contractor shall submit, a revision to the Construction Schedule. Contractor may also request reasonable revisions to currently accepted Construction Schedule in event the Contractor's planning for the work is revised. If Contractor desires to make changes, Contractor shall notify Contracting Officer in writing, stating reason for proposed revision. Accepted revisions shall be incorporated into currently accepted Construction Schedule for next monthly schedule update.
  2. Procedure: If revision to currently accepted Construction Schedule is contemplated, Contractor or Contracting Officer shall advise the other in writing at least seven calendar days prior to next monthly schedule update meeting, describing revision and reasons for the revision. Government-requested revisions will be presented in writing to the Contractor, who shall respond in writing within seven calendar days.



3.2 TIME IMPACT ANALYSIS FOR CONTRACT MODIFICATIONS CHANGES DELAYS AND CONTRACTOR REQUESTS:

1. Requirements: When contract modifications or changes are initiated, delays experienced, or Contractor desires to revise currently accepted Construction Schedule, Contractor shall submit to Contracting Officer a written time impact analysis illustrating the influence of modification, change, delay, or Contractor request on contract time.
2. Time Extensions: Activity delays, resulting in a late completion date projection, shall not automatically mean an extension of contract time is warranted or due to Contractor. It is possible a modification, change, or delay will not affect existing critical path activities or cause non-critical activities to become critical. A modification, change, or delay may result in absorbing a part of available total float that may exist within an activity chain of the Schedule, not causing any effect on contract time. Time extensions will be granted in accordance with terms of contract.
3. Extension of contract time will be granted only to the extent the equitable time adjustments to activity or activities affected by modification, change, or delay exceeds total (positive or zero) float available on a particular activity.
4. Procedure: Each time impact analysis shall be submitted within time period stated in a request for proposal, or time period designated under the clauses entitled Changes or Default. In cases where Contractor does not submit a written request for extension of time and a time impact analysis within the designated time, it is mutually agreed that the particular modification, change, delay, or Contractor request does not require an extension of the contract time. Upon acceptance, time impact analysis shall be incorporated into currently accepted Construction Schedule at next monthly schedule update.
5. Contract Modifications: Prepare time-impact analysis using fragnets to demonstrate effect of proposed change on overall Construction Schedule for each proposed contract modification concurrent with submission.

**END OF SECTION 01 32 16**

**SECTION 01 32 33 – PHOTO DOCUMENTATION**

**PART 1 – GENERAL**

1.1 SUMMARY

- A. Section includes administrative and procedural requirements for:
  1. Existing condition images
  2. Periodic construction images
- B. See Section 01 77 00 "Closeout Procedures" for a complete listing of closeout documents.

## 1.2 SUBMITTALS

- A. Construction Images: Submit images electronically within 7 (seven) days. Include:
  - 1. Date, time and number (sequentially number all images) in filename.
  - 2. Description of vantage point, indicating location, direction (by compass point), and elevation or story of construction.
  - 3. Submit digital images exactly as originally recorded in digital camera, without alteration, or modifications using image-editing software.
- B. Closeout: Submit complete set of digital image electronic files as a Project Record Document.

## PART 2 – PRODUCTS

### 2.1 FORMAT REQUIREMENTS

- A. Images: Provide sRGB (standard Red Green Blue) color images in JPEG (Joint Photographic Experts Group) format. Minimum sensor size of 8 megapixels, and at image resolution of not less than 1600 by 1200, and 300 dpi (dots per inch).

## PART 3 – EXECUTION

### 3.1 CONSTRUCTION IMAGES

- A. General: Take digital images using the maximum range of depth of field, in-focus, to clearly show the Work. No blurry or out-of-focus areas accepted.
  - 1. Maintain index with each set of Construction images and identify the number, date, time, and description for each.
  - 2. Maintain electronic set of images accessible in field at Project site available for reference.
- B. Existing Condition Images: Before starting, take color digital images of Project site and surrounding properties, including existing items to remain during construction, from different vantage points.
  - 1. Flag excavation areas and construction limits before recording construction images.
  - 2. Take separate images to show existing conditions adjacent to property before starting Work.
  - 3. Take separate images of existing buildings either on or adjoining property to accurately record physical conditions at start of construction.
- C. Periodic Construction Images: Take intermittent color, digital images showing construction progress, documenting new work or existing conditions before being covered or concealed, documenting pertinent conditions related to Application for Payment or as requested by the

Contracting Officer. Select vantage points to show status of construction and progress since last images were taken.

- D. Additional Images: Contracting Officer may issue requests for additional images.
  - 1. Three days advance, where feasible.
  - 2. In emergency situations, take additional images within 24 hours of request.

## **END OF SECTION 01 32 33**

## **SECTION 01 33 23 - SUBMITTAL PROCEDURES**

### **PART 1 – GENERAL**

#### **1.1 SUMMARY**

- A. Section includes administrative and procedural requirements for submitting Shop Drawings, Product Data, Samples, and other submittals.

#### **1.2 DEFINITIONS**

- A. Action Submittals: Written, graphic information, and physical samples that require Government's responsive action.
- B. Informational Submittals: Written information that does not require Government's responsive action. Submittals may be rejected for not complying with requirements.
- C. Portable Document Format (PDF): An open standard file format licensed by Adobe Systems used for representing documents in a device-independent and display resolution-independent fixed-layout document format.

#### **1.3 GENERAL SUBMITTAL PROCEDURES**

- A. General: Prepare and submit submittals required by individual Specification Sections and in some cases as requested in drawings. Types of submittals are indicated in individual specific sections.
  - 1. Contracting Officer (CO) reserves right to require submittals in addition to those called for in individual sections.
- B. Coordination: Coordinate preparation and processing of submittals with performance of construction activities. Review for legibility, accuracy, completeness, and compliance with Contract Documents.
  - 1. Coordinate submittal with fabrication, purchasing, testing, delivery, other submittals, and related activities that require sequential activity.

2. Coordinate transmittal of different types of submittals for related parts of Work so processing will not be delayed because of need for concurrent review coordination.
  - a. Contracting Officer reserves right to withhold action on a submittal requiring coordination with other submittals until related submittals are received.
- C. Submittal List: Submittal list is attached to the end of this Specification Section. The intent is to provide an overall summary of submittal requirements. The requirements of individual Specification Sections and terms and conditions of the Contract still apply regardless of what is shown on submittal list.
- D. Processing Time: Allow time for submittal review, including time for re-submittals, as follows. Time for review shall commence when e-mail notification is received by Contracting Officer (or designee) indicating submittal has been posted on NPS management software website and is ready for review. When Contracting Officer has completed review, e-mail notification will be sent to Contractor indicating submittal has been processed. No extension of Contract Time will be authorized because of failure to transmit submittals in advance of Work to permit processing, including re-submittals.
  1. Action Submittals
    - a. Initial Review: Allow 15 days for initial review of each submittal. Allow additional time if coordination with subsequent submittals is required.
    - b. Re-submittal Review: Allow 15 days for review of each re-submittal.
  2. Informational submittals
    - a. Review: Allow 10 days for review of each submittal.
- E. Approved Equals:
  1. For each item proposed as an "approved equal," submit supporting data, including:
    - a. Drawings and samples as appropriate.
    - b. Comparison of the characteristics of the proposed item with that specified.
    - c. Changes required in other elements of the work because of the substitution.
    - d. Name, address, and telephone number of vendor.
    - e. Manufacturer's literature regarding installation, operation, and maintenance, including schematics for electrical and hydraulic systems, lubrication requirements, and parts lists. Describe availability of maintenance service, and state source of replacement materials.
  2. A request for approval constitutes a representation that Contractor:
    - a. Has investigated the proposed item and determined that it is equal or superior in all respects to that specified.
    - b. Will provide the same warranties for the proposed item as for the item specified.
    - c. Has determined that the proposed item is compatible with interfacing items.

- d. Will coordinate installation of an approved item and make changes required in other elements of the work because of the substitution.
  - e. Waives claims for additional expenses that may be incurred as a result of the substitution.
- F. Electronic Submittals: Identify and incorporate information in each electronic submittal file as follows:
  - 1. Transmittal Form (CM-16): All submittals shall be transmitted using National Park Service Transmittal Form (CM-16). The form can be downloaded from the DSC Workflows website's Submittal Review page. No action will be taken on a submittal item unless accompanied by this Transmittal Form.
    - a. Complete the general information at the top of form.
    - b. Provide all required information based on submittal type
    - c. Attach all related documents.
    - d. Sign the Contractor section at bottom of the Transmittal Form (CM-16).
  - 2. Physical samples: Complete Transmittal Form (CM-16) as described above. Deliver physical sample to the Contracting Officer (or designee) on site for processing. All comments and actions will be documented on the Transmittal Form (CM-16).
- G. Identification: Submittal number or other unique identifier, including revision identifier.
  - 1. Submittal number shall use a sequential number (e.g. .001). Re-submittals shall include alphabetic suffix after another decimal point (e.g. .001.A).
- H. Re-submittals: Make re-submittals using same process used with initial submittal.
  - 1. Note date and content of previous submittal.
  - 2. Note date and content of revision in the title block on the Transmittal Form (CM-16) and clearly indicate extent of revision.
  - 3. Re-submit submittals until they are marked "Approved" or "Approved with notations".
- I. Distribution: Furnish copies of final submittals to manufacturers, subcontractors, suppliers, fabricators, installers, and others as necessary for performance of construction activities.
- J. Use for Construction: Use only final submittals with mark indicating "Approved" or "Approved with notations". Ensure notations have been incorporated and, at a minimum, keep one copy of final approved submittal on site for use during construction.

## PART 2 – PRODUCTS

### 2.1 ACTION SUBMITTALS

- A. Product Data: Collect information into a single submittal for each element of construction and type of product or equipment.

1. If information must be specially prepared for submittal because standard printed data are not suitable for use, submit as Shop Drawings, not as Product Data.
  2. Mark each submittal to show which products and options are applicable.
  3. As applicable, include:
    - a. Manufacturer's product specifications.
    - b. Manufacturer's installation instructions: When Contract Documents require compliance with manufacturer's printed instructions, provide one complete set of instructions to Contracting Officer and keep another complete set of instructions at the project site until substantial completion.
    - c. Manufacturer's catalog cuts: Submit only pertinent pages; mark each page of standard printed data to identify specific products proposed for use.
    - d. Wiring diagrams showing factory-installed wiring.
    - e. Printed performance curves.
    - f. Operational range diagrams.
    - g. Compliance with specified referenced standards.
    - h. Testing by recognized testing agency.
  4. Submit product data in PDF (portable document format) file format before or concurrent with samples.
- B. Shop Drawings: Prepare Project-specific information, drawn accurately to scale. Do not base Shop Drawings on reproductions of the Contract Documents or standard printed data.
1. Preparation: Fully illustrate requirements in Contract Documents. As applicable, include:
    - a. Dimensions
    - b. Identification of products
    - c. Fabrication and installation drawings
    - d. Roughing-in and setting diagrams
    - e. Wiring diagrams showing field-installed wiring, including power, signal, and control wiring
    - f. Shopwork manufacturing instructions
    - g. Templates and patterns
    - h. Schedules
    - i. Notation of coordination requirements
    - j. Notation of dimensions established by field measurement
    - k. Relationship to adjoining construction clearly indicated
    - l. Seal and signature of professional engineer if specified
    - m. Wiring Diagrams: Differentiate between manufacturer-installed and field-installed wiring
  2. Submit shop drawings as PDF electronic file
- C. Samples: Submit Samples for review of kind, color, pattern, and texture for a check of these characteristics with other elements and for a comparison of these characteristics between submittal and actual component as delivered and installed.
- D. Construction Materials: Contractor is encouraged to submit products made out of recycled or environmentally responsible material. Every effort will be made by National Park Service to approve these materials.

## 2.2 INFORMATIONAL SUBMITTALS

- A. General: Prepare and submit Informational Submittals required by individual Specification Sections.
  - 1. Certificates and Certifications: Provide a notarized statement with signature of entity responsible for preparing certification. Certificates and certifications shall be signed by officer or other individual authorized to sign documents on behalf of that entity.
  - 2. Informational submittals that do not comply with requirements specified in Contract Documents will be rejected.
- B. Coordination Drawings: Comply with requirements specified in Section 01 31 00 "Project Management and Coordination."
- C. Contractors Construction Schedule: Comply with requirements specified in Section 01 32 16 "Construction Schedule."
- D. Accident Prevention Plan: Comply with requirements specified in Section 01 35 23 "Safety Requirements."
- E. Schedule of Values: Comply with requirements specified in Section 01 32 16 "Construction Schedule."
- F. Waste Recycling Plan: Comply with requirements specified in Section 01 74 19 "Construction Waste Management and Disposal."
- G. Quality Control Plan: Comply with requirements specified in Section 01 40 00 "Quality Requirements."
- H. Pedestrian/Vehicular Traffic Control Plan: Comply with requirements specified in Section 01 50 00, Part 3.3, C "Temporary Facilities and Controls, Support Facilities Installation, Traffic Controls."
- I. Qualification Data: Prepare written information demonstrating capabilities and experience of firm or person. Include lists of completed projects with project names and addresses, names and addresses of architects and owners, and other information specified.
- J. Installer Certificates: Prepare written statements on manufacturer's letterhead certifying Installer complies with Contract Documents and, where required, is authorized by manufacturer for this specific Project.
- K. Manufacturer Certificates: Prepare written statements on manufacturer's letterhead certifying manufacturer complies with Contract Documents. Include evidence of manufacturing experience where required.
- L. Product Certificates: Prepare written statements on manufacturer's letterhead certifying product complies with Contract Documents.
- M. Material Certificates: Prepare written statements on manufacturer's letterhead certifying material complies with Contract Documents.

- N. Product Test Reports: Prepare written reports indicating current product produced by manufacturer complies with requirements in Contract Documents. Base reports on evaluation of tests performed by manufacturer and witnessed by a qualified testing agency, or on comprehensive tests performed by a qualified testing agency.
- O. Research/Evaluation Reports: Prepare written evidence, from a model code organization acceptable to authorities having jurisdiction, that product complies with building code in effect for Project.
- P. Preconstruction Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of tests performed before installation of product, for compliance with performance requirements in Contract Documents.
- Q. Compatibility Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of compatibility tests performed before installation of product. Include written recommendations for primers and substrate preparation needed for adhesion.
- R. Design Data: Prepare written and graphic information, including: performance and design criteria, list of applicable codes and regulations, and calculations. Include list of assumptions and other performance and design criteria and a summary of loads. Include load diagrams if applicable. Provide name and version of software, if any, used for calculations. Include page numbers.
- S. Manufacturer's Instructions: Prepare written or published information documenting manufacturer's recommendations, guidelines, and procedures for installing or operating a product or equipment. Include name of product and name, address, and telephone number of manufacturer.
- T. Permit Compliance Products: Prepare required information for compliance with permit provisions. Products include written notification of project startup, suspension, and completion of work; photo documentation of site conditions; reports; and drawings.

### PART 3 – EXECUTION

#### 3.1 CONTRACTOR'S REVIEW

- A. Review each submittal and check for coordination with other Work of Contract and for compliance with Contract Documents. Note corrections and field dimensions.

#### 3.2 CONTRACTING OFFICER'S ACTION

- A. General: Submittals will be disapproved without technical review if identification information is missing, not filled in, or if placed on back of submittal; an incorrect format of submittals is provided; transmittal form is incorrectly filled out; submittals are not coordinated; or submittals do not show evidence of Contractor's approval.



1. Any work done or orders for materials or services placed before approval shall be at Contractor's own risk.
- B. Action Submittals: Contracting Officer will review each submittal, generate comments on corrections or modifications required, and indicate appropriate action on the Transmittal Form (CM-16). Submittal will be marked as defined below:
  1. APPROVED: Acceptable with no corrections.
  2. APPROVED WITH NOTATIONS: Minor corrections or clarifications required. Comments are clear and no further review is required. Contractor shall address review comments when proceeding with the work.
  3. DISAPPROVED - RESUBMIT: Rejected as not in accordance with the contract or as requiring major corrections or clarifications. Contracting Officer will identify reasons for disapproval. Contractor shall revise and resubmit with changes clearly identified.
- C. Informational Submittals: Contracting Officer will review each submittal and will either accept or reject it.
- D. Partial submittals are not acceptable, will be considered non-responsive, and will be returned without review.

## **END OF SECTION 01 33 23**

## **SECTION 01 35 23 - SAFETY REQUIREMENTS**

### **GENERAL**

#### **1.1 SUMMARY**

- A. Section includes establishing an effective accident prevention program and providing a safe working environment for personnel and visitors.

#### **1.2 CONDITIONS PRESENT FOR PROJECT**

- A. Shoreline activities will encompass the following conditions: Drilling, hydro-jetting, trenching, excavation, digging, crane and lifting equipment procedures, wildlife, weather, drowning, flooding, turbulent water, fatigue, exposure to wind, rain, heat, humidity, cold temperatures and cold water exposure.

#### **1.3 SUBMITTALS**

- A. Site-Specific Accident Prevention Plan (APP): Submit APP after contract award and before Pre-Construction conference. Contracting Officer (CO) will review proposed APP. If APP requires any revisions or corrections, Contractor shall resubmit Plan within 10 days. No progress payments will be made until the APP is accepted.

1. The Site-Specific Accident Prevention Plan is not the Company Safety Program but is a plan to address hazards, work conditions, and abatement measures specific to the contract.

#### 1.4 QUALITY ASSURANCE

- A. Comply with contract clauses "Accident Prevention" and "Permits and Responsibilities." In case of conflicts between Federal, State, and local safety and health requirements, the most stringent shall apply. Onsite equipment shall meet 29 CFR 1926 (Code of Federal Regulations) (Occupational Safety and Health Administration (OSHA)) requirements. Failure to comply with requirements of this section and related sections may result in suspension of work.
- B. Site Safety Supervisor:
  1. Designate authorized onsite representative for preparation and maintenance of the APP.
  2. Shall be responsible for:
    - a. Implementation and enforcement of the APP
    - b. Daily safety inspections
    - c. Conducting and documenting weekly and monthly safety meetings
    - d. Review of safety requirements at progress meetings
    - e. Compilation and maintenance of Safety Data Sheets (SDS) and safety reference materials
    - f. Tracking and resolution of safety violations
    - g. Site personnel and visitor compliance with site safety and health requirements and APP
    - h. Investigation and reporting of accidents and injuries
- C. Qualifications of Employees:
  1. Physically and able to perform their assigned duties in a safe manner.
  2. Do not allow employees whose ability or alertness is impaired because of prescription or illegal drug use, fatigue, illness, intoxication, or other conditions that may expose themselves or others to injury to perform work.
  3. Provide operating instructions for equipment. Operators of vehicles, hoisting equipment, and hazardous plant equipment shall be able to understand signs, signals, operating instructions, and be fully capable of operating such equipment. Retain copies of operator licenses and certifications onsite.

#### 1.5 ACCIDENT REPORTING

- A. Reportable Accidents: Defined as: death, occupational disease, and/or traumatic injury to employees or the public; fires; and/or property damage by accident in excess of \$100.
  1. Notify Contracting Officer immediately in the event of a reportable accident.

2. Fill out and forward an Accident/Property Damage Report Form (CM-22) to Contracting Officer within 7 days of a reportable accident. Obtain form from Contracting Officer.

## PART 2 – PRODUCTS

### 2.1 ACCIDENT PREVENTION PLAN (APP)

- A. APP shall be written to comply with OSHA and project requirements (generic plan is not acceptable) including but not limited to:
  1. Name and qualifications of supervisor responsible to carry out program.
  2. Weekly and monthly safety meetings shall be documented with topics and attendees.
  3. First aid and rescue procedures.
  4. Job Hazard Analysis (JHA) for each major phase. List of hazards associated and methods proposed to provide for property protection and safety of the public, National Park Service personnel, and Contractor's employees. Include initial and continuing training.
    - a. The contractor shall specify methods chosen for hazard control/abatement. For example: will the contractor use personal fall arrest, guard rails, or safety nets to protect employees from fall hazards. The contractor should supply enough detail to allow reviewers to evaluate.
  5. Planning for possible emergency situations, as detailed in Article 1.2. Such planning shall take nature of construction, site conditions, and degree of exposure of persons and property into consideration.

### 2.2 FIRST AID FACILITIES

- A. Provide adequate facilities for number of employees and appropriate to construction hazards.

### 2.3 PERSONNEL PROTECTIVE EQUIPMENT (PPE)

- A. Selection shall comply with 29CFR 1910 Subpart I and 19CFR1910.120.

## PART 3 – EXECUTION

### 3.1 DAILY SAFETY INSPECTIONS

- A. Conduct daily safety inspections and maintain daily safety reports which include:
  1. Area/operation inspected
  2. Date of inspection
  3. Identified hazards
  4. Corrective actions taken

### 3.2 EMERGENCY INSTRUCTIONS

- A. Post telephone numbers and reporting instructions for ambulance, physician, hospital, fire department, and police in conspicuous locations at work site.

### 3.3 FIRE AND LIFE SAFETY

- A. Comply with requirements of National Fire Protection Association (NFPA) 241 (Standard for Safeguarding Construction, Alteration, and Demolition Operations).

### 3.4 HAZARDOUS MATERIALS

- A. Hazardous materials: Explosive, flammable, poisonous, corrosive, oxidizing, irritating, or otherwise harmful substances that could cause death or injury. Hazardous materials include pesticides, biocides, and carcinogens as listed by recognized authorities, such as the Environmental Protection Agency (EPA), the International Agency for Research on Cancer (IARC), or 29CFR1010 Subpart Z.
- B. Store hazardous materials in accordance with manufacturer's and OSHA Subpart H requirements. Maintain Safety Data Sheets (SDS) for each chemical readily available on site.
  - 1. Immediately report spills of hazardous materials to the Park.
  - 2. Maintain a spill emergency response kit.
  - 3. Train employees how to respond to a spill and use emergency response kit.

### 3.5 PROTECTIVE EQUIPMENT

- A. Inspect personal protective equipment daily and maintain in a serviceable condition. Clean, sanitize, and repair personal items as appropriate before issuing to another individual.

### 3.6 SAFETY MEETINGS

- A. As a minimum, conduct one weekly 15 -minute "toolbox" safety meeting conducted by a foreman or supervisor and attended by construction personnel at worksite. Topics shall coincide with work scheduled for following week. Document and submit meeting minutes to Contracting Officer within one day after meeting.
- B. Conduct monthly safety meetings for personnel, contractors, and subcontractors performing work on the site. Notify Contracting Officer of meeting dates and times. Meetings shall be used to: review effectiveness of Contractor's safety effort; resolve current health and safety problems; provide a forum for planning safe construction activities, and for updating Accident Prevention Plan. Contracting Officers Representative will attend meetings and enter results of meetings into the daily log.

### 3.7 HARD HATS AND PROTECTIVE EQUIPMENT AREAS

- A. A hard hat use area shall be designated by Contractor. Hard hat area shall be posted by Contractor in a manner satisfactory to Contracting Officer.
- B. It is Contractor's responsibility to require persons working on or visiting site to wear hard hats and PPE in good repair at all times. As a minimum, maintain [six] <Insert number> hard hats and other APP required equipment.

### 3.8 TRAINING

- A. First Aid: Provide training to personnel to ensure prompt and efficient first aid.
- B. Hazardous Material: Train and instruct each employee exposed to hazardous material in safe and approved methods of handling and storage.
- C. Park Provided Training: Parks may require and provide specific mandatory training for all on-site construction personnel related to employee and visitor safety, Park rules and law enforcement, or wildlife and resource protection.

## END OF SECTION 01 35 23

## SECTION 01 40 00 - QUALITY REQUIREMENTS

### PART 1 – GENERAL

#### 1.1 SUMMARY

- A. Section includes administrative and procedural requirements for quality assurance and quality control.
- B. Testing and inspecting services are required to verify compliance with requirements specified or indicated. These services do not relieve Contractor of responsibility for compliance with Contract Document requirements. Quality of work shall be responsibility of the Contractor.
  - 1. Specified tests, inspections, and related actions do not limit Contractor's other quality assurance and control procedures that facilitate compliance with Contract Document requirements.
- C. See Divisions 2 through 49 Sections for specific test and inspection requirements.

#### 1.2 DEFINITIONS

- A. Quality Assurance Services: Activities, actions, and procedures performed before and during execution of the work to guard against defects and deficiencies and substantiate that proposed construction will comply with requirements.

- B. Quality Control Services: Tests, inspections, procedures, and related actions during and after execution of the work to evaluate actual products incorporated into the work and completed construction comply with requirements.
- C. Preconstruction Testing: Tests and inspections performed specifically for project before products and materials are incorporated into work to verify performance or compliance with specified criteria.
- D. Product Testing: Tests and inspections performed by a Nationally Recognized Testing Laboratory (NRTL), a National Voluntary Laboratory Accreditation Program (NVLAP), or a testing agency qualified to conduct product testing, to establish product performance and compliance with industry standards.
- E. Source Quality Control Testing: Tests and inspections performed at the source, i.e., plant, mill, factory, or shop.
- F. Field Quality Control Testing: Tests and inspections performed on-site for installation of work and for completed work.
- G. Testing Agency or Laboratory: Entity engaged to perform specific tests, inspections, or both. Testing laboratory shall mean the same as testing agency.
- H. Installer/Applicator/Erector: Contractor or another entity engaged by Contractor as an employee, Subcontractor, or Sub-subcontractor, to perform a particular construction operation, including installation, erection, application, and similar operations.
  - 1. Using a term such as "carpentry" does not imply certain construction activities must be performed by accredited or unionized individuals of a corresponding generic name, such as "carpenter." It also does not imply that requirements specified apply exclusively to trades people of corresponding generic name.

### 1.3 CONFLICTING REQUIREMENTS

- A. "Reference Standards" Paragraph below may resolve issues that sometimes arise when using referenced standards due to conflicting requirements in the Contract Documents. Coordinate with Conditions of the Contract. Revise to suit Project, office policy, and National Park Service (NPS) requirements.
- B. Reference Standards: If compliance with two or more standards is specified and standards establish different or conflicting requirements for minimum quality levels, comply with most stringent requirement. Refer uncertainties and requirements that are different, but apparently equal, to Contracting Officer before proceeding.
- C. Minimum Quality Levels: Quality level shown or specified shall be minimum provided or performed. Actual installation may comply exactly with minimum quality specified, or it may exceed minimum within reasonable limits. To comply with requirements, indicated numeric values are minimum or maximum, as appropriate, for the context of requirements. Refer uncertainties to Contracting Officer before proceeding.

#### 1.4 SUBMITTALS

##### A. Quality Control Plan:

1. After contract award and before Pre-Construction conference, submit a written Contractor Quality Control (CQC) plan.
2. If plan requires revisions or corrections, Contractor shall resubmit plan within 10 days.
3. Government reserves the right to require changes in plan during contract period as necessary to obtain the quality specified.
4. No change in the approved plan may be made without written concurrence by Contracting Officer.

##### B. Qualification Data: For testing agencies specified in "Quality Assurance" Article to demonstrate their capabilities and experience. Include proof of qualifications in form of a recent report on inspection of testing agency by a recognized authority.

##### C. Contractor Quality Control (CQC) Daily Reports: Submit showing inspections and tests on first workday following date covered by report. (Download from DSC Workflows website > Forms/Templates/Samples/Guidelines page > Construction Forms section.)

1. Review Construction Management Representative (CMR) Daily report if applicable and reconcile any differences prior to posting.

##### D. Test Reports

1. Test reports shall be completed by person performing test.
2. Submit Daily Test Information Sheets with Quality Control Daily Reports.
3. Submit failing test results and proposed remedial actions within four hours of noted deficiency.
4. Submit three copies of complete test results no later than one calendar day after test was performed.

##### E. Accessibility Inspection Report:

1. Fill out applicable sections of the Accessibility Inspection Report and attach to Contractor Quality Control Daily Report.
2. Utilize attached Accessibility Inspection form to document compliance with Architectural Barriers Act Accessibility Standards (ABAAS).
3. Inspect at various stages of construction as needed to ensure finished product meets standards.
4. Submit report no later than one calendar day after inspection was performed.

##### F. Off-Site Inspection Reports: Submit prior to shipment.

##### G. If Contractor Quality Control plan and Quality Control Daily Reports are not submitted as specified, Contracting Officer may retain payments until such time plan(s) is/are accepted and implemented, or may retain payments for work completed on days with no Quality Control Daily Reports.

- H. Permits, Licenses, and Certificates: For National Park Service (NPS) records, submit copies of permits, licenses, certifications, inspection reports, releases, jurisdictional settlements, notices, receipts for fee payments, judgments, correspondence, records, and similar documents, established for compliance with standards and regulations bearing on performance of work.

## 1.5 QUALITY ASSURANCE

- A. General: Qualifications paragraphs in this Article establish minimum qualification levels required; individual Specification Sections specify additional requirements.
- B. Contractors Quality Control Staff:
  - 1. Contractor's Quality Control Supervisor may also perform other duties.
  - 2. Contractor's designated Quality Control Supervisor shall be on the project site whenever contract work is in progress.
  - 3. Contractor's job supervisory staff may be used to assist Quality Control Supervisor supplemented, as necessary, by additional certified testing technicians.
- C. Installer Qualifications: Firm or individual experienced in installing, erecting, or assembling work similar in material, design, and extent indicated for Project, whose work has resulted in construction with a record of successful in-service performance.
- D. Manufacturer Qualifications: Firm experienced in manufacturing products or systems similar to those indicated for Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.
- E. Fabricator Qualifications: Firm experienced in producing products similar to those indicated for Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.
- F. Professional Engineer Qualifications: Professional engineer legally qualified to practice in jurisdiction where Project is located and is experienced in providing engineering services of kind indicated (including Structural Tests and Special Inspections (STSI)). Engineering services are defined as those performed for installations of system, assembly, or products similar to those indicated for Project in material, design, and extent.
- G. Specialists: Certain sections of the Specifications require that specific construction activities shall be performed by entities who are recognized experts in those operations. Specialists shall satisfy qualification requirements indicated and shall be engaged for activities indicated.
  - 1. Requirement for specialists shall not supersede building codes and regulations governing Work.
- H. Testing Agency Qualifications: A Nationally Recognized Testing Laboratory (NRTL), a National Voluntary Laboratory Accreditation Program (NVLAP), or an independent agency with experience and capability to conduct testing and inspecting indicated, according to ASTM E 329; and with additional qualifications specified in individual Sections; and where required by Contract, is acceptable to Contracting Officer.



1. Nationally Recognized Testing Laboratory (NRTL): Nationally recognized testing laboratory according to 29 CFR 1910.7 (Code of Federal Regulations).
  2. National Voluntary Laboratory Accreditation Program (NVLAP): Testing agency accredited according to National Institute of Standards and Technology's (NIST) National Voluntary Laboratory Accreditation Program.
  3. Measuring devices, laboratory equipment, and instruments shall be calibrated at established intervals against certified standards in accordance with NIST requirements. Measuring and testing devices shall be made available for use by Government for verification tests.
- I. Factory-Authorized Service Representative Qualifications: Authorized representative of manufacturer who is trained and approved by manufacturer to inspect installation of manufacturer's products similar in material, design, and extent to those indicated for Project.

## 1.6 QUALITY CONTROL

- A. Contractor is responsible for testing and inspections, including Structural Tests and Special Inspections (STSI), as identified in attached STSI. Inspect and test work as needed to ensure quality of materials, workmanship, construction, finish, and functional performance are in compliance with applicable specifications, drawings, and those required by the Building Code.
1. Engage qualified testing agency to perform quality-control services.
  2. Submit appropriate report for each quality-control service.
  3. Testing and inspecting requested by Contractor and not required by Contract Documents are Contractor's responsibility.
  4. Contracting Officer may designate test locations.
- B. Manufacturer's Field Services: Where indicated, engage factory-authorized service representative to inspect field-assembled components and equipment installation, including service connections. Report results in writing.
- C. Re-testing/Re-inspecting: Regardless of whether original tests or inspections were Contractor's responsibility, provide quality-control services, including retesting and re-inspecting, for construction of replaced work that failed to comply with Contract Documents.
- D. Testing Agency Responsibilities: Cooperate with NPS and Contractor in performance of duties. Provide qualified personnel to perform required tests and inspections.
1. Notify Contracting Officer and Contractor promptly of irregularities or deficiencies observed in work during performance of services.
  2. Determine location from which test samples will be taken and in which in-situ tests are conducted.
  3. Conduct and interpret tests and inspections, State in each report whether tested and inspected work complies with or deviates from requirements.
  4. Submit a certified report of each test, inspection, and similar quality-control service through Contractor.
  5. Do not release, revoke, alter, or increase Contract Document requirements or approve or accept any portion of Work.

- E. Associated Services: Cooperate with agencies performing required tests, inspections, and similar quality-control services, and provide reasonable auxiliary services as requested. Notify agency sufficiently in advance of operations to permit assignment of personnel. Provide:
1. Access to Work.
  2. Incidental labor and facilities necessary to facilitate tests and inspections.
  3. Adequate quantities of representative samples of materials that require testing and inspecting. Assist agency in obtaining samples.
  4. Facilities for storage and field curing of test samples.
  5. Delivery of samples to testing agencies.
  6. Preliminary design mix proposed for material mixes that require control by testing agency.
  7. Security and protection for samples and testing and inspecting equipment at Project site.
- F. Coordination: Coordinate sequence of activities to accommodate required quality assurance and control services with minimum delay and to avoid removing and replacing construction to accommodate testing and inspecting.
1. Schedule times for tests, inspections, obtaining samples, and similar activities.

## PART 2 – PRODUCTS

### 2.1 QUALITY CONTROL PLAN

- A. Quality Control Plan shall include:
1. List of personnel responsible for quality control and assigned duties. Include each person's qualifications. Include alternate(s) and qualifications.
  2. Copy of letter of direction to Contractor's Quality Control Supervisor(s) outlining assigned duties and authorities designated by principal or owner.
  3. Names, qualifications / accreditations, and descriptions of laboratories to perform sampling and testing, and samples of proposed report forms from laboratories.
  4. Methods of performing, documenting, and enforcing quality control of work including Contractor report forms and acknowledgment of NPS forms.
  5. Methods of monitoring and controlling environmental pollution and contamination as required by regulations and laws.
  6. Specific discussion regarding mockups, off-site visits, receiving inspections, manufacturers representation, startup requirements, and other aspects of performance specific to Project.
  7. Provisions for substantial completion(s) and final inspection(s) per Contract.

## PART 3 – EXECUTION

### 3.1 OFF-SITE CONTROL

- A. Items fabricated or assembled off-site shall be inspected for quality control at place of fabrication.

### 3.2 ON-SITE CONTROL

- A. Notification:

- 1. Notify Contracting Officer at least 48 hours in advance of preparatory phase meeting.
  - 2. Notify Contracting Officer at least 24 hours in advance of initial and follow-up phases.

- B. Preparatory Phase: Perform before beginning each feature of work.

- 1. Review control submittal requirements with personnel directly responsible for quality assurance and quantity control of the work. As a minimum, Contractor's Quality Control Supervisor and foreman responsible for feature of work shall be in attendance.
  - 2. Review applicable specifications sections and drawings related to feature of work.
  - 3. Ensure copies of referenced standards related to sampling, testing, and execution for feature of work are available on site.
  - 4. Ensure provisions have been made for field control testing.
  - 5. Examine work area to ensure preliminary work has been completed.
  - 6. Verify field dimensions and advise Contracting Officer of discrepancies with contract documents.
  - 7. Ensure necessary equipment and materials are at project site and they comply with approved shop drawings and submittals.
  - 8. Document preparatory phase activities and discussions on Contractor's Quality Control Daily Report.

- C. Initial Phase:

- 1. As soon as work begins, inspect and test representative portion of particular feature of work for quality of workmanship.
  - 2. Review control testing procedures to ensure compliance with contract requirements.
  - 3. Document initial phase activities and discussions on Contractor's Quality Control Daily Report. Exact location of initial phase shall be indicated for future reference and comparison with follow-up phases.

- D. Follow-Up Phase: Inspect and test as work progresses to ensure compliance with contract requirements until completion of work.

- E. Additional Preparatory and Initial Phases: Additional preparatory and initial phases may be required on same feature of work for following reasons:

1. Quality of on-going work is unacceptable.
2. Changes in quality control staff, on-site production supervision, or work crew.
3. Work on particular feature of work is resumed after substantial period of inactivity.

### 3.3 DOCUMENTATION

- A. Maintain Quality Control Daily Reports, Daily Test Report Information Sheets, and Accessibility Inspection Reports of quality control activities and tests. (Download from DSC Workflows website > Forms/Templates/Samples/Guidelines page > Construction Forms section.)
- B. Quality Control Daily Reports do not fulfill the written notification requirements of contract clauses, such as Disputes, Differing Site Conditions, or Changes, and will not be accepted as a substitute notification.
- C. Quality Control Daily Reports shall accurately portray all work and materials. The materials shall match the materials as approved from the Construction Documents (CD) requirements. Combustible/flammable materials shall be documented as well.

### 3.4 ENFORCEMENT

- A. Contractor shall stop work on any item or feature pending satisfactory correction of deficiency noted by quality control staff or Contracting Officer.

### 3.5 REPAIR AND PROTECTION

- A. General: On completion of testing, inspecting, sample taking, and similar services, repair damaged construction and restore substrates and finishes.
  1. Provide materials and comply with installation requirements specified in other Specification Sections. Restore patched areas and extend restoration into adjoining areas with durable seams as invisible as possible.
- B. Protect construction exposed by or for quality-control service activities.
- C. Repair and protection are Contractor's responsibility, regardless of the assignment of responsibility for quality-control services.

**END OF SECTION 01 40 00**

## **SECTION 01 50 00 - TEMPORARY FACILITIES AND CONTROLS**

### **PART 1 – GENERAL**

#### **1.1 SUMMARY**

- A. Section includes requirements for temporary utilities, support facilities, and security and protection facilities.

#### **1.2 DEFINITIONS – Not Used**

#### **1.3 USE CHARGES – Not Used**

#### **1.4 QUALITY ASSURANCE**

- B. Environmental Protection: Provide environmental protection as required by agency(ies) with jurisdiction and as indicated in Contract Documents. Coordinate with requirements of the following:
  - 1. Regulatory Requirements
  - 2. Indoor Air Quality (IAQ) Management
  - 3. Noise and Acoustics Management
  - 4. Environmental Management
  - 5. Construction Waste Management
- C. Accessible Temporary Egress: Comply with applicable provisions in the United States (U.S.) Architectural & Transportation Barriers Compliance Board's Architectural Barriers Act Accessibility Standard (ABAAS) Accessibility Guidelines.

### **PART 2 – PRODUCTS**

#### **2.1 MATERIALS**

- A. Temporary materials may be new or used, but must be adequate in capacity for required usage, must not create unsafe conditions, and must not violate requirements of applicable codes and standards.
- B. Safety Barrier Fence: Orange plastic fence, minimum height, 4 feet.
- C. Barrier Tape: Yellow tape Imprinted with "CAUTION: CONSTRUCTION AREA," manufactured by Reef Industries, Inc., Houston, Texas, or approved equal.
- D. Dust-Control Adhesive-Surface Walk-off Mats: Provide mats minimum 36 by 60 inches.

## 2.2 TEMPORARY FACILITIES

- A. Field Offices, General: Prefabricated or mobile units with serviceable finishes, temperature controls, and foundations adequate for normal loading.
- B. Storage and Fabrication Sheds: Temporary weather tight sheds or other covered facilities for storage of materials subject to weather damage. Number and size of structures shall be subject to Contracting Officer's approval.
- C. Toilets: Sufficiently lighted and ventilated toilet facilities in weatherproof, sight proof, handicap accessible, sturdy enclosures with privacy locks.
  - 1. Provide with hand washing/cleaning station and/or dispenser.

## 2.3 EQUIPMENT

- A. Fire Extinguishers: Portable, UL rated; with class and extinguishing agent as required by locations and classes of fire exposures.

## PART 3 – EXECUTION

### 3.1 INSTALLATION, GENERAL

- A. Locate facilities where they will serve Project adequately and result in minimum interference with performance of the Work. Relocate and modify facilities as required by progress of the Work.
  - 1. Locate facilities to limit site disturbance and as directed by the Contracting Officer.
- B. Provide each facility ready for use when needed to avoid delay. Do not remove until facilities are no longer needed or are replaced by authorized use of completed permanent facilities.

### 3.2 TEMPORARY UTILITY INSTALLATION

- A. General: Install temporary service or connect to existing service.
  - 1. Arrange with utility company, NPS, and existing users for time when service can be interrupted, if necessary, to make connections for temporary services. Acquire necessary permits.
- B. Non-potable water for construction is not available within the park boundaries. Contractor shall furnish non-potable water from a source outside park boundary.
- C. Potable water is not available on site. Furnish cool, potable water for construction personnel in locations convenient to work stations.
- D. Sanitary Facilities: Provide temporary toilets, and wash facilities for use by construction personnel.

1. Place in approved location. Relocate as work progress requires.
2. Maintain and clean toilet facilities at least weekly.
3. Completely remove sanitary facilities on completion of work.
4. Toilets: Use of existing toilet facilities will be permitted, as long as facilities are cleaned and maintained in a condition acceptable to NPS. At Substantial Completion, restore facilities to condition existing before initial use.

### 3.3 SUPPORT FACILITIES INSTALLATION

#### A. General: Comply with the following:

1. Provide incombustible construction for offices, shops, and sheds located within construction area or within 50 feet of building lines. Comply with NFPA 241.
2. Maintain support facilities until near Substantial Completion. Remove structures, equipment, and furnishings, and terminate services after punch list is 100 percent completed or when directed by Contracting Officer. Personnel remaining after Substantial Completion will be permitted to use permanent facilities, under conditions acceptable to Contracting Officer.

#### B. Traffic Controls: Erect and maintain barricades, lights, danger signals, and warning signs in accordance with Manual on Uniform Traffic Control Devices (MUTCD), Part IV, latest edition.

1. Protect existing site improvements to remain including curbs, pavement, and utilities.
2. Maintain access for fire-fighting equipment and access to fire hydrants.
3. Illuminate barricades and obstructions at night; keep safety lights burning from sunset to sunrise.
4. Adequately barricade and post open cuts in or adjacent to thoroughfares.
5. Protect pedestrian traffic by guardrails or fences.
6. When pedestrian traffic is detoured onto a roadway, provide temporary walkways with protection as required at ends and overhead. For walkways, use lumber running parallel to direction of traffic movement and provide ramps at changes of elevation.
7. Cover pipes, hoses, and power lines crossing sidewalks and walkways with troughs using beveled edge boards.
8. Install Barrier Tape where directed by Contracting Officer. Keep a minimum of two rolls on site.

#### C. Parking: Use designated parking areas for construction personnel.

#### D. Project Identification and Temporary Signs: Provide Project identification and other signs. Fence, barricade, or otherwise block off the immediate work area to prevent unauthorized entry.

1. Provide temporary, directional signs for construction personnel and visitors.
2. Maintain and touchup signs so they are legible at all times.
3. Erect and maintain sufficient detour signs at road closures and along detour routes.

- E. Waste Disposal Facilities: Provide waste-collection containers in sizes adequate to handle waste from construction operations. Comply with requirements of agency(ies) with jurisdiction.
- F. Lifts and Hoists: Provide facilities necessary for hoisting materials and personnel.
  - 1. Truck cranes and similar devices used for hoisting materials are considered "tools and equipment" and not temporary facilities. Individual Project circumstances may require use of other construction aids and miscellaneous facilities, such as scaffolds, platforms, swing stages, ramps and bridges, incidental sheeting and shoring, and demolition waste chutes. Add requirements to suit Project.

### 3.4 SECURITY AND PROTECTION FACILITIES INSTALLATION

- A. Environmental Protection: Provide protection, operate temporary facilities, and conduct construction in ways and by methods that comply with environmental regulations and minimize possible air, waterway, and subsoil contamination or pollution or other undesirable effects.
- B. Cleaning of Equipment: Contractor shall ensure prior to moving on to Project Area, equipment, is free of soil, seeds, vegetative matter, or other debris that could contain or hold seeds. Ensure equipment has been pressure washed and is free of exotic species. Equipment shall be considered free of soil, seeds, and other debris when visual inspection does not disclose such material. Disassembly of equipment components or specialized inspection tools are not required.
- C. Pest Control: Not Used
- D. Site Enclosure Fence: Before construction operations begin, furnish and install fencing to prevent people and animals from easily entering site except by entrance gates.
  - 1. Extent of Fence: As required to enclose entire Project site or portion determined sufficient to accommodate construction operations.
  - 2. Locate vehicular gates to avoid interference with traffic on public thoroughfares.
  - 3. Locate pedestrian entrance gates as required to provide controlled personnel entry.
  - 4. Maintain security by limiting number of keys and restricting distribution to authorized personnel.
- E. Security Enclosure and Lockup: Install substantial temporary enclosure around partially completed areas of construction. Provide lockable entrances to prevent unauthorized entrance, vandalism, theft, and similar violations of security.
- F. Barricades, Warning Signs, and Lights: Comply with requirements of Manual on Uniform Traffic Control Devices (MUTCD), for erecting structurally adequate barricades, including warning signs and lighting.
- G. Temporary Enclosures: Provide temporary enclosures for protection of construction, in progress and completed, from exposure, foul weather, other construction operations, and similar activities. Provide temporary weathertight enclosure for building exterior.



### 3.5 OPERATION, TERMINATION, AND REMOVAL

- A. Supervision: Enforce strict discipline in use of temporary facilities. Minimize waste and abuse, limit availability of temporary facilities to essential and intended uses.
- B. Maintenance: Maintain facilities in good operating condition until removal.
- C. Temporary Facility Changeover: Do not change over from using temporary security and protection facilities to permanent facilities until Substantial Completion.
- D. Termination and Removal: Remove each temporary facility when need for its service has ended, when it has been replaced by authorized use of a permanent facility, or no later than Substantial Completion. Complete or, if necessary, restore permanent construction that may have been delayed because of interference with temporary facility. Repair damaged Work, clean exposed surfaces, and replace construction that cannot be satisfactorily repaired.
  - 1. Materials and facilities that constitute temporary facilities are property of Contractor. NPS reserves right to take possession of Project identification signs.
  - 2. At Substantial Completion, clean and renovate permanent facilities used during construction period.

**END OF SECTION 01 50 00**

## **SECTION 01 67 00 - PRODUCT REQUIREMENTS**

### PART 1 – GENERAL

#### 1.1 SUMMARY

- A. Section includes administrative and procedural requirements for selection of products for use in Project; product delivery, storage, and handling; manufacturers' standard warranties on products; special warranties; and environmental requirements.

#### 1.2 DEFINITIONS

- A. Products: Items purchased for incorporating into Work, whether purchased for Project or taken from previously purchased stock. The term "product" includes "material," "equipment," "system," and terms of similar intent.
  - 1. Named Products: Items identified by manufacturer's product name, including make or model number or other designation shown or listed in manufacturer's published product literature, current as of date of Contract Documents.
  - 2. New Products: Items that have not previously been incorporated into another project or facility, except that products consisting of recycled-content materials are allowed, unless explicitly stated otherwise. Products salvaged or recycled from other projects are not considered new products.

3. Comparable Product: Product demonstrated and approved through submittal process, or where indicated as a product substitution, to have indicated qualities related to type, function, dimension, in-service performance, physical properties, appearance, and other characteristics that equal or exceed those of specified product.
- B. Biobased Materials: As defined in Farm Security and Rural Investment Act, for purposes of Federal procurement of biobased products, "biobased" means a "commercial or industrial product (other than food or feed) composed, in whole or in significant part, of biological products or renewable domestic agricultural materials (including plant, animal, and marine materials) or forestry materials." Biobased materials also include fuels, chemicals, building materials, or electric power or heat produced from biomass as defined by The Biomass Research and Development Act of 2000.
1. Biobased content: Amount of biobased carbon in material or product as a percentage of weight (mass) of total organic carbon in material or product.
- C. Chain-of-Custody: Process whereby a product or material is maintained under physical possession or control during its entire life cycle.
- D. Environmentally preferable products: Products and services with lesser or reduced effect on the environment in comparison to conventional products and services. Refer to Environmental Protection Agency's (EPA) Final Guidance on Environmentally Preferable Purchasing for more information.
- E. Stewardship: Responsible use and management of resources in support of sustainability.
- F. Sustainability: Maintenance of ecosystem components and functions for future generations.
1. Recycled Content Materials: Products containing pre-consumer or post-consumer materials as all or part of their feedstock. Recycled content claim shall be consistent with International Organization for Standardization (ISO) 140001 Standard for the Use of Environmental Marketing Claims.
  2. Rapidly Renewable Material: Material made from plants typically harvested within a ten-year cycle.
  3. Regional Materials: Materials manufactured and extracted, harvested, or recovered within a radius of 500 miles from Project location.

### 1.3 SUBMITTALS

- A. Record Submittals as specified in – Sustainable Design Close-Out Documentation, submit:
1. Material Safety Data Sheets (MSDS): For each product required by OSHA to have a MSDS, submit an MSDS. MSDS shall be prepared within the previous five years. Include information for MSDS Sections 1 to 16 in accordance with ANSI Z400.1 and as follows:
    - a. Section 1: Chemical Product and Company Identification
    - b. Section 2: Composition/Information on Ingredients
    - c. Section 3: Hazards Identification
    - d. Section 4: First Aid Measures

- e. Section 5: Fire Fighting Measures
- f. Section 6: Accidental Release Measures
- g. Section 7: Handling and Storage
- h. Section 8: Exposure Controls/Person Protection
- i. Section 9: Physical and Chemical Properties
- j. Section 10: Stability and Reactivity Data
- k. Section 11: Toxicological Information. Include data used to determine the hazards cited in Section 3. Identify acute data, carcinogenicity, reproductive effects, and target organ effects
- l. Section 12: Ecological Information. Include data regarding environmental impacts during raw materials acquisition, manufacture, and use. Include data regarding environmental impacts in event of accidental release.
- m. Section 13: Disposal Considerations. Include data regarding proper disposal of the chemical. Include information regarding recycling and reuse. Indicate whether or not product is considered to be "hazardous waste" according to United States EPA Hazardous Waste Regulations 40 CFR 261 (Code of Federal Regulations).
- n. Section 14: Transportation Information. Identify hazard class for shipping.
- o. Section 15: Regulatory Information. Identify federal, state, and local regulations applicable to the material.
- p. Section 16: Other Information. Include additional information relative to recycled content, biobased content, and other information regarding environmental and health impacts.

#### 1.4 QUALITY ASSURANCE

- A. Compatibility of Options: If Contractor is given option of selecting between two or more products for use on Project, product selected shall be compatible with products previously selected, even if previously selected products were also options.

#### 1.5 PRODUCT DELIVERY, STORAGE, AND HANDLING

- A. Deliver, store, and handle products using means and methods that prevent damage, deterioration, and loss, including theft. Comply with manufacturer's written instructions.
- B. Delivery and Handling:
  - 1. Schedule delivery to minimize long-term storage at Project site and prevent overcrowding of construction spaces.
  - 2. Coordinate delivery with installation time to ensure minimum holding time for items that are flammable, hazardous, easily damaged, or sensitive to deterioration, theft, and other losses.
  - 3. Deliver products to Project site in undamaged condition; in manufacturer's original sealed container or other packaging system; complete with labels and instructions for handling, storing, unpacking, protecting, and installing.
  - 4. Inspect products on delivery to ensure compliance with Contract Documents. Ensure products are undamaged and properly protected.
  - 5. Obtain materials in biodegradable or recyclable/reusable packaging which uses minimum amount of packaging possible.

C. Storage:

1. Allow for inspection and measurement of quantity or counting of units.
2. Store materials in manner to not endanger Project structure.
3. Store products subject to damage by the elements, under cover in weather tight enclosure above ground, with ventilation adequate to prevent condensation.
4. Store cementitious products and materials on elevated platforms.
5. Store foam plastic from exposure to sunlight, except to extent necessary for period of installation and concealment.
6. Comply with product manufacturer's written instructions for temperature, humidity, ventilation, and weather-protection requirements for storage.
7. Protect stored products from damage and liquids from freezing.
8. Store loose granular materials in well-drained area on solid surfaces to prevent mixing with foreign matter.

1.6 PACKAGING

- A. Where Contractor has option to provide one of listed products or equal, preference shall be given to products with minimal packaging and easily recyclable packaging as defined in ASTM D5834.
- B. Maximize use of source reduction and recycling procedures outlined in ASTM D5834.

1.7 ENVIRONMENTALLY PREFERABLE PRODUCTS

- A. Provide environmentally preferable products to greatest extent possible.
1. To greatest extent possible, provide products and materials with a lesser or reduced effect on the environment considering raw materials acquisition, production, manufacturing, packaging, distribution, reuse, operation, maintenance, and/or disposal of the product.
  2. Eliminate use of ozone depleting compounds during and after construction where alternative environmentally preferable products are available, consistent with either Montreal Protocol and Title VI or Clean Air Act Amendments of 1990, or equivalent overall air quality benefits that take into account life cycle impacts.
  3. Use products meeting or exceeding EPA's recycled content recommendations for EPA-designated products. Use materials with recycled content such that the sum of post-consumer recycled content plus one-half of the pre-consumer content constitutes at least 10% (based on cost) of total value of the materials in project.

1.8 PRODUCT WARRANTIES

- A. Warranties specified in other Sections shall be in addition to, and run concurrent with, other warranties required by Contract Documents. Manufacturer's disclaimers and limitations on product warranties do not relieve Contractor of obligations under requirements of Contract Documents.
1. Manufacturer's Warranty: Preprinted written warranty published by individual manufacturer for product specifically endorsed by manufacturer to Owner.

2. Special Warranty: Written warranty required by or incorporated into Contract Documents, either to extend time limit provided by manufacturer's warranty or to provide more rights for Owner.
- B. Special Warranties: Prepare written document containing appropriate terms and identification, ready for execution. Submit draft for approval before final execution.
1. Manufacturer's Standard Form: Modified to include Project-specific information and properly executed.
  2. Specified Form: When specified forms are included with Specifications, prepare written document using appropriate form properly executed.
  3. Refer to Divisions 2 through 49 Sections for specific content requirements and particular requirements for submitting special warranties.
- C. Submittal Time: Comply with requirements in Section 01 77 00 "Closeout Procedures."

## PART 2 – PRODUCTS

### 2.1 PRODUCT SELECTION PROCEDURES

- A. General Product Requirements: Provide products to comply with Contract Documents, undamaged and, unless otherwise indicated, new at time of installation.
1. Provide products complete with accessories, trim, finish, fasteners, and other items needed for a complete installation and indicated use and effect.
  2. Standard Products: If available, and unless custom products or nonstandard options are specified, provide standard products of types produced and used successfully in similar situations on other projects.
  3. Government reserves right to limit selection to products with warranties not in conflict with requirements of Contract Documents.
  4. Where products are accompanied by term "as selected," Contracting Officer will make selection.
  5. Where products are accompanied by term "match sample," sample to be matched is Government's.
  6. Descriptive, performance, and reference standard requirements in Specifications establish "salient characteristics" of products.
- B. Product Selection Procedures:
1. Product: Where Specifications name single product and manufacturer, provide named product that complies with requirements or approved equal.
  2. Manufacturer/Source: Where Specifications name single manufacturer or source, provide product by named manufacturer or source that complies with requirements or approved equal.
  3. Products: Where Specifications include list of names of both products and manufacturers, provide one of the products listed that complies with requirements or approved equal.

4. Manufacturers: Where Specifications include list of manufacturers' names, provide a product by one of the manufacturers listed that complies with requirements or approved equal.
5. Available Products: Where Specifications include list of names of both products and manufacturers, provide one of the products listed, or an unnamed product, that complies with requirements. Comply with provisions in Part 2 "Comparable Products" Article for consideration of an unnamed product.
6. Available Manufacturers: Where Specifications include list of manufacturers, provide a product by one of the manufacturers listed, or an unnamed manufacturer, that complies with requirements. Comply with provisions in Part 2 "Comparable Products" Article for consideration of an unnamed product.
7. Product Options: Where Specifications indicate sizes, profiles, and dimensional requirements on Drawings are based on a specific product or system, provide specified product, system, or approved equal.
8. Basis-of-Design Product: Where Specifications name product and include a list of manufacturers, provide specified product or a comparable product by one of the other named manufacturers, or approved equal. Drawings and Specifications indicate sizes, profiles, dimensions, and other characteristics based on the product named.
9. Visual Matching Specification: Where Specifications require matching an established Sample, select product that complies with requirements and matches the sample. Contracting Officers decision will be final on whether a proposed product matches.
  - a. If no product available within specified category matches and complies with other specified requirements, comply with provisions in Part 2 "Product Substitutions" Article for proposal of product.
10. Visual Selection Specification: Where Specifications include phrase "as selected from manufacturer's colors, patterns, textures" or a similar phrase, select a product that complies with other specified requirements.
  - a. Standard Range: Where Specifications include phrase "standard range of colors, patterns, textures" or similar phrase, Contracting Officer will select color, pattern, density, or texture from manufacturer's product line that does not include premium items.
  - b. Full Range: Where Specifications include phrase "full range of colors, patterns, textures" or similar phrase, Contracting Officer will select color, pattern, density, or texture from manufacturer's product line that includes both standard and premium items.

## 2.2 COMPARABLE PRODUCTS

- A. Conditions: Contracting Officer will consider Contractor's request for comparable product when the following conditions are satisfied. If following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:

1. Evidence proposed product does not require revisions to Contract Documents, that it is consistent with Contract Documents and will produce indicated results and is compatible with other portions of Work.
2. Detailed comparison of significant qualities of proposed product with those named in Specifications. Significant qualities include attributes such as performance, weight, size, durability, visual effect, and specific features and requirements indicated.
3. Evidence proposed product provides specified warranty.
4. List of similar installations for completed projects with project names and addresses and names and addresses of architects and owners, if requested.
5. Samples, if requested.

### **PART 3 – EXECUTION**

#### **3.1 PROTECTION AFTER INSTALLATION**

- A. Provide adequate coverings as necessary to protect installed materials from damage resulting from natural elements, traffic, and subsequent construction. Remove when no longer needed.

### **END OF SECTION 01 67 00**

### **SECTION 01 73 40 - EXECUTION**

#### **PART 1 – GENERAL**

##### **1.1 SUMMARY**

- A. Section includes general procedural requirements governing execution of Work including:
  1. Construction layout
  2. Field engineering and surveying
  3. General installation of products
  4. Protection of installed construction
  5. Correction of the Work

##### **1.2 SUBMITTALS**

- A. Certificates: Submit certificate signed by land surveyor or professional engineer certifying location and elevation of improvements comply with requirements.

### 1.3 QUALITY ASSURANCE

- A. Land Surveyor Qualifications: Professional engineer or land surveyor legally qualified to practice in jurisdiction where Project is located and—is experienced in providing land-surveying services of kind indicated.

### PART 2 – PRODUCTS (Not Used)

### PART 3 – EXECUTION

#### 3.1 EXAMINATION

- A. Existing Conditions: Existence and location of site improvements and other construction indicated as existing are not guaranteed.
  - 1. Before beginning work, investigate and verify the existence and location of other construction affecting the Work.
  - 2. Before construction, verify the location and points of connection of utility services.
- B. Existing Utilities: Not Used.
- C. Acceptance of Conditions: Examine substrates, areas, and conditions, with Installer or Applicator present where indicated, for compliance with requirements for installation tolerances and other conditions affecting performance. Record observations.
  - 1. Proceed with installation only after unsatisfactory conditions have been corrected. Proceeding with the Work indicates acceptance of surfaces and conditions.

#### 3.2 COORDINATION WITH UTILITY SERVICE PROVIDERS – Not Used

#### 3.3 PREPARATION

- A. Field Measurements: Take field measurements as required to fit Work properly. Recheck measurements before installing each product. Where portions of Work are indicated fit to other construction, verify dimensions of other construction by field measurements before fabrication. Coordinate fabrication schedule with construction progress to avoid delaying Work.
- B. Space Requirements: Verify space requirements and dimensions of items shown diagrammatically on Drawings.
- C. Review of Contract Documents and Field Conditions: Immediately on discovery of need for clarification of the Contract Documents caused by differing field conditions outside control of Contractor, submit request for information to Contracting Officer in accordance with Section 01 31 00 "Project Management and Coordination."



### 3.4 CONSTRUCTION LAYOUT

- A. Verification: Verify layout information shown on Drawings, in relation to the existing benchmarks before proceeding to lay out Work. Notify Contracting Officer promptly if discrepancies are discovered.
- B. General: Engage a land surveyor or professional engineer to lay out Work using accepted surveying practices.
  - 1. Establish benchmarks and control points to set lines of construction and elsewhere as needed to locate each element of Project.
  - 2. Establish dimensions within tolerances indicated. Do not scale Drawings to obtain required dimensions.
  - 3. Inform installers of lines and levels to which they must comply.
  - 4. Check location, level and plumb, of every major element as Work progresses.
  - 5. Notify Contracting Officer when deviations from required lines and levels exceed allowable tolerances.
- C. Site Improvements: Locate and lay out site improvements, including grading, fill and topsoil placement, slopes, and invert elevations.
- D. Building Lines and Levels: Locate and lay out control lines and levels for structures. Transfer survey markings and elevations for use with control lines and levels. Level foundations and piers from two or more locations.
- E. Record Log: Maintain log of layout control work. Record deviations from required lines and levels. Include beginning and ending dates and times of surveys, weather conditions, name and duty of each survey party member, and types of instruments and tapes used. Make log available for review by National Park Service (NPS).

### 3.5 FIELD ENGINEERING

- A. Reference Points: Locate existing permanent benchmarks, control points, and similar reference points before beginning Work. Preserve and protect permanent benchmarks and control points during construction operations. Controls destroyed by Contractor will be replaced by Contractor at their expense.
  - 1. Existing Monuments: All benchmarks, land corners, and triangulation points, established by other surveys, existing within construction area shall be preserved. If existing monuments interfere with Work, secure written permission before removing them.
- B. Benchmarks: Establish and maintain a minimum of two (2) permanent benchmarks on Project site, referenced to data established by survey control points. Comply with NPS requirements for type and size of benchmark.
  - 1. Record benchmark locations, with horizontal and vertical data, on Project Record Documents.

- C. Certified Survey: On completion of foundation walls, major site improvements, and other work requiring field-engineering services, prepare a certified survey showing dimensions, locations, angles, and elevations of construction and sitework.

### 3.6 INSTALLATION

- A. General: Locate Work and components of Work accurately in correct alignment and elevation, as indicated.
  - 1. Make vertical work plumb and make horizontal work level.
  - 2. Where space is limited, install components to maximize space available for maintenance and ease of removal for replacement.
- B. Comply with manufacturer's written instructions and recommendations for installing products in applications indicated.
- C. Install products at the time and under conditions for best possible results. Maintain conditions required for product performance until Substantial Completion.
- D. Conduct construction operations so no part of the Work is subjected to damaging operations or loading in excess of that expected during normal conditions.
- E. Tools and Equipment: Use appropriate tools and recommended equipment per manufacturer instructions to accomplish work.
- F. Templates: Obtain and distribute to parties involved templates for work specified to be factory prepared and field installed. Check Shop Drawings of other work to confirm adequate provisions are made for locating and installing products to comply with indicated requirements.
- G. Hazardous Materials: Use products, cleaners, and installation materials not considered hazardous.

### 3.7 PROGRESS CLEANING

- A. General: Clean Project site and work areas daily. Coordinate progress cleaning for joint-use areas where more than one Installer has worked. Enforce requirements strictly. Dispose of materials lawfully.
- B. Site: Maintain Project site free of waste materials and debris.
- C. Installed Work: Keep installed work clean. Clean installed surfaces according to written instructions of manufacturer or fabricator of product installed, using only cleaning materials specifically recommended. If specific cleaning materials are not recommended, use cleaning materials not hazardous to health or property and will not damage exposed surfaces.
- D. Exposed Surfaces in Finished Areas: Clean exposed surfaces and protect from damage and deterioration at time of Substantial Completion.

- E. Waste Disposal: Do not bury or burn waste materials on-site. Do not wash waste materials down sewers, into waterways or into ocean.
- F. Clean and protect construction in progress and adjoining materials already in place during handling and installation. Apply protective covering where required to ensure protection from damage or deterioration at Substantial Completion.
- G. Limiting Exposures: Supervise construction operations so that no part of construction completed or in progress, is subject to harmful, dangerous, damaging, or deleterious exposure during construction period.
- H. Final Cleaning: At completion of Work, remove remaining waste materials, rubbish, tools, equipment, machinery and surplus materials. Clean exposed surfaces and leave Project clean and ready for occupancy.
  - 1. Provide final cleaning in accordance with ASTM E1971.

### 3.8 PROTECTION OF INSTALLED CONSTRUCTION

- A. Provide final protection and maintain conditions that ensure installed Work is without damage or deterioration at time of Substantial Completion.
- B. Comply with manufacturer's written instructions for temperature and relative humidity.

### 3.9 CORRECTION OF THE WORK

- A. Repair or remove and replace defective construction. Restore damaged substrates and finishes.
  - 1. Repairing includes replacing defective parts, refinishing damaged surfaces, touching up with matching materials, and properly adjusting operating equipment.
- B. Restore permanent facilities used during construction to specified condition.
- C. Remove and replace damaged surfaces exposed to view if surfaces cannot be repaired without visible evidence of repair.
- D. Repair components that do not operate properly. Remove and replace operating components that cannot be repaired.
- E. Remove and replace chipped, scratched, and broken glass or reflective surfaces.

**END OF SECTION 01 73 40**

## **SECTION 01 74 19 - CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL**

### **PART 1 – GENERAL**

#### **1.1 SUMMARY**

- A. Section includes administrative and procedural requirements for:
  - 1. Disposing of nonhazardous demolition and construction waste.

#### **1.2 DEFINITIONS**

- A. Construction Waste: Building and site improvement materials and other solid waste resulting from construction, remodeling, renovation, or repair operations. Construction waste includes packaging.
- B. Demolition Waste: Building and site improvement materials resulting from demolition or selective demolition operations.
- C. Debris: Non-hazardous solid waste generated during construction, demolition, or renovation of a structure which exceeds 2.5-inch particle size that is: a manufactured object; plant or animal matter; or natural geologic material (e.g., cobbles and boulders). A mixture of debris and other material such as soil or sludge is also subject to regulation as debris if mixture is comprised primarily of debris by volume, based on visual inspection.
- D. Disposal: Removal off-site of demolition and construction waste and subsequent sale, recycling, reuse, or deposit in landfill or incinerator acceptable to authorities having jurisdiction.
- E. Environmental Pollution and Damage: Presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances; or degrade utility of environment for aesthetic, cultural, or historical purposes.
- F. Garbage: Refuse and scraps resulting from preparation, cooking, dispensing, and consumption of food.
- G. Hazardous Materials: Material regulated as a hazardous material in accordance with 49 CFR 173 (Code of Federal Regulations), requires a Material Safety Data Sheet (MSDS) in accordance with 29 CFR 1910.1200, or which during end use, treatment, handling, storage, transportation or disposal meets or has components which meet or have potential to meet the definition of Hazardous Waste in accordance with 40 CFR 261.
- H. Recycle: Recovery of demolition or construction waste for subsequent processing in preparation for reuse.
- I. Salvage: Recovery of demolition or construction waste and subsequent sale or reuse in another facility.

- J. Single-use plastic products: Plastic items intended to be disposed of immediately after use, including plastic and polystyrene food and beverage containers, bottles, straws, cups, cutlery, and disposable plastic bags.

### 1.3 PERFORMANCE REQUIREMENTS

- A. General: Project shall minimize creation of construction, deconstruction, and demolition waste to protect and restore natural habitat and resources. Minimize factors contributing to waste such as over packaging, improper storage, ordering error, poor planning, breakage, mishandling, single-use plastic products, and contamination. A Waste Management Plan shall be developed to ensure that existing site and building materials are reused, salvaged, or recycled. Minimize waste disposal in landfills.
- B. If waste materials encountered during deconstruction/demolition or construction phase are found to contain lead, asbestos, polychlorinated biphenyls (PCBs), (such as fluorescent lamp ballasts), or other harmful substances, they are to be handled and removed in accordance with local, state, and federal laws and requirements concerning hazardous waste.

### 1.4 SUBMITTALS

- A. Waste Management Plan: After award of contract and prior to scheduled Pre-Construction Conference, Contractor shall submit a draft Waste Management Plan to Contracting Officer for approval. Revise and resubmit Plan as required by Contracting Officer. Approval of Contractor's Plan will not relieve Contractor of responsibility for compliance with applicable environmental regulations.
- B. Records of Donations: Indicate receipt and acceptance of salvageable waste donated to individuals and organizations. Indicate whether organization is tax exempt.
- C. Records of Sales: Indicate receipt and acceptance of salvageable waste sold to individuals and organizations. Indicate whether organization is tax exempt.
- D. Recycling and Processing Facility Records: Indicate receipt and acceptance of recyclable waste by recycling and processing facilities licensed to accept them. Include manifests, weight tickets, receipts, and invoices.
- E. Landfill and Incinerator Disposal Records: Indicate receipt and acceptance of waste by landfills and incinerator facilities licensed to accept them. Include manifests, weight tickets, receipts, and invoices.
- F. Statement of Refrigerant Recovery: Signed by qualified refrigerant recovery technician responsible for recovering refrigerant, stating refrigerant that was present was recovered and recovery was performed according to Environmental Protection Agency (EPA) regulations. Include name and address of technician and date refrigerant was recovered.
- G. Closeout Submittals
  - 1. With Closeout Submittals, submit a summary of a Project Waste Management Plan worksheet for solid waste disposal and diversion.

## 1.5 QUALITY ASSURANCE

- A. Regulatory Requirements: Comply with hauling and disposal regulations of authorities having jurisdiction.
- B. Waste Management Meeting: Conduct separate meeting or cover in Pre-Construction Conference and comply with requirements in Section 01 31 00 "Project Management and Coordination." Review methods and procedures related to waste management including:
  - 1. Review and discuss waste management plan.
  - 2. Review requirements for documenting quantities of each type of waste and its disposition.
  - 3. Review and finalize procedures for materials separation and verify availability of containers and bins needed to avoid delays.
  - 4. Review procedures for periodic waste collection and transportation to recycling and disposal facilities.
  - 5. Review procedures for reduction of single-use plastic products on site.
  - 6. Review waste management requirements for each trade.

## PART 2 – PRODUCTS

### 2.1 WASTE MANAGEMENT PLAN

- A. General: Develop plan consisting of waste identification and waste reduction work plan. Include separate sections in plan for demolition and construction waste. Indicate quantities by weight or volume but use same units of measure throughout waste management plan.
- B. Waste Identification: Indicate anticipated types and quantities of demolition, site-clearing and construction waste generated by the Work. Include estimated quantities and assumptions for estimates.

## PART 3 – EXECUTION

### 3.1 PLAN IMPLEMENTATION

- A. General: Implement waste management plan as approved by Contracting Officer. Provide handling, containers, storage, signage, transportation, and other items as required to implement waste management plan during entire duration of Contract.
- B. Contractor shall establish contacts with local recycling and reuse companies to set up lines of responsibility. Contractor shall be responsible for coordination in terms of identifying materials, pickup schedules, and standard quality for recycled materials.
- C. Training: Review the waste management plan with and train workers, subcontractors, and suppliers on proper waste management procedures, as appropriate for the Work occurring at Project site.

- D. Site Access and Temporary Controls: Conduct waste management operations to ensure minimum interference with roads, streets, walks, walkways, and other adjacent occupied and used facilities.
- E. Materials handling procedures: Material to be recycled shall be protected from contamination and shall be handled, stored, and transported in a manner that meets requirements set by designated facilities for acceptance. Establish defined area for operations of each trade, especially woodcutting so off-cuts are kept in one area and can be sorted by dimension for future reuse.

### 3.2 SALVAGING DEMOLITION WASTE (Not Used)

### 3.3 RECYCLING DEMOLITION AND CONSTRUCTION WASTE, GENERAL

- A. General: Recycle paper and beverage containers used by on-site workers.
- B. Procedures: Separate recyclable waste from other waste materials, trash, and debris. Separate recyclable waste by type at Project site to maximum extent practical.
  - 1. Provide appropriately marked containers or bins for controlling recyclable waste until removed from Project site. Include list of acceptable and unacceptable materials at each container and bin.
    - a. Inspect containers and bins for contamination and remove contaminated materials if found.
  - 2. Stockpile processed materials on-site without intermixing with other materials. Place, grade, and shape stockpiles to drain surface water. Cover to prevent windblown dust.
  - 3. Stockpile materials away from construction area. Do not store within drip line of trees.
  - 4. Store components off ground and protect from weather.
  - 5. Remove recyclable waste off Government property and transport to recycling receiver or processor.

### 3.4 RECYCLING DEMOLITION WASTE (Not Used)

### 3.5 RECYCLING CONSTRUCTION WASTE (Not Used)

### 3.6 DISPOSAL OF WASTE

- A. General: Except for items or materials to be salvaged, recycled, or otherwise reused, remove waste materials from Project site and legally dispose in landfill or incinerator acceptable to authorities having jurisdiction.
  - 1. Except as otherwise specified, do not allow waste materials to accumulate on-site.

2. Remove and transport debris in manner preventing spillage on adjacent surfaces and areas.
- B. Burning: Do not burn waste materials.
  - C. Burning: Burning of waste materials allowed only at designated areas on Government property, provided required permits are obtained. Provide full-time monitoring for burning materials until fires are extinguished.
  - D. Disposal: Transport waste materials off Government property and legally dispose of them.

## **END OF SECTION 01 74 19**

## **SECTION 01 77 00 - CLOSEOUT PROCEDURES**

### **PART 1 – GENERAL**

#### **1.1 SUMMARY**

- A. Section includes administrative and procedural requirements for contract closeout, including:
  1. Project Record Drawings
  2. Closeout Submittals
  3. Substantial Completion and Final Inspection
  4. Permit Closure and Transfer
  5. Final Acceptance of the Work
  6. Warranties

#### **1.2 PROJECT RECORD DRAWINGS**

- A. Maintain one complete full-size set of contract drawings and one full-size set of vendor-supplied drawings. Clearly mark changes, deletions, and additions using National Park Service (NPS) drafting standards to show actual construction conditions. Show additions in red, deletions in green and special instructions in blue.
- B. Keep record drawings current. Make record drawings available to Contracting Officer (CO) for inspection at the time of monthly progress payment requests. If project record drawings are not current, Contracting Officer may retain an appropriate amount of progress payment.
- C. Submit complete record drawings on completion of total project. Include shop drawings, sketches, and additional drawings to be included in final set, with clear instructions showing the location of these drawings.



### 1.3 CLOSEOUT SUBMITTALS

- D. A list of closeout requirements has been attached at the end of the Division 1 Specifications for your convenience. The intent is to provide an overall summary of requirements and not a comprehensive list. Terms and conditions of the contract require satisfaction of requirements of individual specification sections regardless of what is shown on the list. Submit the following before requesting final inspection:
1. Specific warranties, guarantees, workmanship bonds, final certifications, and similar documents.
  2. Project Record Documents, final completion construction digital images recorded in PDF, with index and descriptions, and similar final record information.
  3. Deliver tools, spare parts, extra materials, and similar items to location designated by Contracting Officer. Label with manufacturer's name and model number where applicable.
    - a. Special Tools: One set of special tools required to operate, adjust, dismantle, or repair equipment. Special tools are those not normally found in possession of mechanics or maintenance personnel.
  4. Make final changeover of permanent locks and deliver keys to Contracting Officer. Advise Park personnel of changeover in security provisions.
  5. Terminate and remove temporary facilities, mockups, construction tools, and similar elements from Project site, complete final cleaning requirements, including touchup painting.
  6. Touch up and repair and restore marred exposed finishes to eliminate visual defects.
  7. Instruct NPS personnel in operation, adjustment, and maintenance of products, equipment, and systems.

### 1.4 FINAL INSPECTION, SUBSTANTIAL COMPLETION AND ACCEPTANCE

- A. Request final inspection in writing when project or designated portion of project is substantially complete. Contracting Officer will proceed with inspection within 10 days of receipt of written request or will advise Contractor of items that prevent project from being substantially complete.
- B. If work is determined substantially complete, following final inspection, Contracting Officer will prepare Punch List and issue a Letter of Substantial Completion.
- C. If work is not determined substantially complete following final inspection, Contracting Officer will notify Contractor in writing. Contractor shall request new final inspection after completing work. Re-inspection costs may be charged against Contractor in accordance with Inspection of Construction contract clause.
- D. Contractor shall complete Punch List within 30 calendar days, documented weather permitting.
- E. If Contractor completes items of work on Punch List and contractually required items, Contracting Officer will issue Letter of final acceptance of work.

- F. If Contractor fails to complete work within the time frame, Contracting Officer may correct work with an appropriate reduction in contract price or charge for re-inspection costs in accordance with Inspection of Construction contract clause.

## 1.5 PERMIT CLOSURE AND TRANSFER

- A. After substantial completion and Punch List completion, permits shall be closed and documented by Agency(ies) with Jurisdiction for the permit.

## 1.6 WARRANTIES

- B. Submittal Time: Submit written warranties on request of Contracting Officer for designated portions of Work where commencement of warranties other than date of Substantial Completion is indicated.
- C. Organize warranty documents into an orderly sequence based on table of contents of the Operations and Maintenance Manual.
  - 1. Warranty Electronic File: Scan warranties and bonds and assemble complete warranty and bond submittal package into a single indexed electronic PDF (portable document format) file with links enabling navigation to each item. Provide bookmarked table of contents at beginning of document.
- D. Provide copies of each warranty document in the operation and maintenance manuals.

## PART 2 – PRODUCTS

### 2.1 MATERIALS

- A. See Division 1 Specification Section "Execution" for information on cleaning agents.

## PART 3 – EXECUTION

### 3.1 FINAL CLEANING

- A. General: Conduct final cleaning and waste-removal operations to comply with local laws and ordinances and Federal and local environmental and antipollution regulations.
- B. Cleaning: Employ experienced workers or professional cleaners for final cleaning. Clean each surface or unit to condition expected in an average commercial building cleaning and maintenance program. Comply with manufacturer's written instructions.
  - 1. Complete following cleaning operations before requesting inspection for certification of Substantial Completion for entire Project or for a portion of Project:

- a. Clean Project site, yard, and grounds, in areas disturbed by construction activities, including landscape development areas, of rubbish, waste material, litter, and other foreign substances.
  - b. Sweep paved areas broom clean. Remove petrochemical spills, stains, and other foreign deposits.
  - c. Rake grounds that are neither planted nor paved to smooth, even-textured surface.
  - d. Remove tools, construction equipment, machinery, and surplus material from Project site.
  - e. Touch up, repair, and restore marred, exposed finishes and surfaces. Replace finishes and surfaces that cannot be satisfactorily repaired or restored or that already show evidence of repair or restoration.
- C. Waste Disposal: Comply with requirements of Section 01 74 19 "Construction Waste Management and Disposal."

## **END OF SECTION 01 77 00**

## **SECTION 02 41 19 - SELECTIVE DEMOLITION**

### **PART 1 – GENERAL**

#### **1.1 SUMMARY**

##### **A. The Work of this Section Includes:**

- 1. Demolition and removal of selected portions of site elements.

##### **B. Related Requirements:**

- 1. Section 011000 "Summary of Work " for restrictions on use of the premises, Owner-occupancy requirements, and phasing requirements.

#### **1.2 DEFINITIONS**

- A. Remove: Detach items from existing construction and legally dispose of off-site unless indicated to be removed and salvaged or removed and reinstalled.
- B. Existing to Remain: Existing items of construction that are not to be removed.

#### **1.3 MATERIALS OWNERSHIP**

- A. Unless otherwise indicated, demolition waste becomes property of Contractor.

- B. Historic items, relics, antiques, and similar objects including, but not limited to, cornerstones and their contents, commemorative plaques and tablets, and other items of interest or value to Owner that may be uncovered during demolition remain the property of Owner.

- 1. Carefully salvage in a manner to prevent damage and promptly return to Owner.

#### 1.4 COORDINATION

- A. Arrange selective demolition schedule so as not to interfere with Owner's operations.

#### 1.5 PREINSTALLATION MEETINGS – Not Used

#### 1.6 INFORMATIONAL SUBMITTALS

- A. Survey of Existing Conditions: Submit survey.
- B. Proposed Protection Measures: Submit report, including Drawings, that indicates the measures proposed for protecting individuals and property. Indicate proposed locations and construction of barriers.
- C. "Schedule of Selective Demolition Activities" Paragraph below may be used to track Contractor's progress; it may also be used to determine that selective demolition will not interfere with Owner's operations.
- D. Warranties: Documentation indicating that existing warranties are still in effect after completion of selective demolition.

#### 1.7 CLOSEOUT SUBMITTALS

- A. Inventory: Submit a list of items that have been removed.

#### 1.8 QUALITY ASSURANCE – Not Used

#### 1.9 FIELD CONDITIONS

- A. Owner will not occupy area immediately adjacent to selective demolition area. Conduct selective demolition so Owner's operations will not be disrupted.
- B. Notify Architect of discrepancies between existing conditions and Drawings before proceeding with selective demolition.
- C. Hazardous Materials: It is not expected that hazardous materials will be encountered in the Work.

1. If materials suspected of containing hazardous materials are encountered, do not disturb; immediately notify Architect and Owner. Hazardous materials will be removed by Owner under a separate contract.
- D. Required Mitigations: For the proposed project actions to be within compliance requirements during construction and/or project implementation, the following mitigations must be adhered to:
1. In the event that human remains are discovered during maintenance activities, all work on the project must stop and the park archeologist contacted immediately. As required by law, the coroner will be notified first. All provisions outlined in the Native American Graves Protection and Repatriation Act (1990) will be followed.
  2. If unknown paleontological resources are discovered during construction, work in that location would be stopped until the resources could be properly recorded and evaluated. Measures would be taken to avoid further resource impacts or to mitigate their loss or disturbance
  3. Although there is no surface evidence of archeological resources, clearance to proceed is recommended with the condition that if concealed archeological resources are encountered during project activities, all necessary steps will be taken to protect them, and the Park Cultural Resources Manager will be notified immediately.
- E. On-site sale of removed items or materials is not permitted.

## PART 2 - PRODUCTS

### 2.1 PERFORMANCE REQUIREMENTS

- A. Regulatory Requirements: Comply with governing EPA notification regulations before beginning selective demolition. Comply with hauling and disposal regulations of authorities having jurisdiction.
- B. Standards: Comply with ANSI/ASSP A10.6 and NFPA 241.

## PART 3 - EXECUTION

### 3.1 EXAMINATION

- A. Review Project Record Documents of existing construction or other existing condition and hazardous material information provided by Owner. Owner does not guarantee that existing conditions are same as those indicated in Project Record Documents.
- B. Survey of Existing Conditions: Record existing conditions by use of preconstruction photographs or video. Comply with Section 013233 "Photographic Documentation."

1. Inventory and record the condition of items to be removed. Photograph or video conditions that might be misconstrued as damage caused by removal.
2. Photograph or video existing conditions of adjoining construction including finish surfaces, that might be misconstrued as damage caused by selective demolition operations.

### 3.2 PREPARATION

- A. Temporary Shoring: Design, provide, and maintain shoring, bracing, and structural supports as required to preserve stability and prevent movement, settlement, or collapse of construction and finishes to remain, and to prevent unexpected or uncontrolled movement or collapse of construction being demolished.
  1. Strengthen or add new supports when required during progress of selective demolition.
- B. Temporary Protection: Provide temporary barricades and other protection required to prevent injury to people and damage to adjacent structures.
- C. Existing Items to Remain: Protect construction indicated to remain against damage and soiling during selective demolition. When permitted by CO, items may be removed to a suitable, protected storage location and reinstalled in their original locations after selective demolition operations are complete.

### 3.3 UTILITY SERVICES AND BUILDING SYSTEMS – Not Used

### 3.4 SALVAGE/REINSTALL – Not Used

### 3.5 SELECTIVE DEMOLITION, GENERAL

- A. General: Demolish and remove existing construction only to extent required by new construction and as indicated. Use methods required to complete the Work within limitations of governing regulations and as follows:
  1. Proceed with selective demolition systematically, from one end to the other.
  2. Use removal methods least likely to damage adjacent construction to remain. Use hand tools or small power tools designed for sawing or grinding, not hammering and chopping. Temporarily cover openings securely to remain.
  3. Remove unsuitable materials and promptly dispose of off-site.
  4. Remove components and lower to ground by method suitable to avoid free fall and to prevent ground impact or debris generation.
  5. Locate selective demolition equipment and remove debris and materials.
- B. Site Access and Temporary Controls: Conduct selective demolition and debris-removal operations to ensure minimum interference with public access roads, walkways, and other adjacent occupied and used facilities.

1. Do not close or obstruct streets, walks, walkways, or other adjacent occupied or used facilities without permission from Owner and authorities having jurisdiction. Provide alternate routes around closed or obstructed trafficways if required by authorities having jurisdiction.
2. Comply with governing environmental-protection regulations.

### 3.6 SELECTIVE DEMOLITION, FOR SPECIFIC MATERIALS – Not Used

### 3.7 DISPOSAL OF DEMOLISHED MATERIALS

- A. Remove demolition waste materials from Project site and dispose of them in an EPA-approved construction and demolition waste landfill acceptable to authorities having jurisdiction. Recycle or dispose of them in accordance with Section 01 74 19 "Construction Waste Management and Disposal."
  1. Do not allow demolished materials to accumulate on-site.
  2. Remove and transport debris in a manner that will prevent spillage on adjacent surfaces and areas.
  3. Comply with requirements specified in Section 017419 "Construction Waste Management and Disposal."
- B. Burning: Do not burn demolished materials.

### 3.8 CLEANING

- A. Clean adjacent structures and areas of debris caused by selective demolition operations. Return adjacent areas to condition existing before selective demolition operations began.

### 3.9 SELECTIVE DEMOLITION SCHEDULE

- A. Remove: Twenty-three (23) damaged bollards.
- B. Existing to Remain: All other undamaged bollards

## **END OF SECTION 02 41 19**

## **SECTION 10 14 23 – PANEL SIGNAGE**

### **PART 1 - GENERAL**

#### **1.1 SUMMARY**

- A. Section Includes:

1. Panel signs.
- 1.2 ALLOWANCES – Not Used
- 1.3 UNIT PRICES – Not Used
- 1.4 DEFINITIONS
  - A. Accessible: In accordance with the accessibility standard.
- 1.5 COORDINATION – Not Used
- 1.6 ACTION SUBMITTALS
  - A. Product Data:
    1. Panel signs.
  - B. Shop Drawings: For panel signs.
    1. Include fabrication and installation details and attachments to other work.
    2. Show sign mounting heights, locations of supplementary supports to be provided by other installers, and accessories.
    3. Show message list, typestyles, graphic elements and layout for each sign.
  - C. Samples for Verification: For each type of sign assembly showing all components and with the required finish(es), in manufacturer's standard size unless otherwise indicated and as follows:
    1. Panel Signs: Full-size Sample.
    2. Full-size Samples, if approved, will be returned to Contractor for use in Project.
  - D. Manufacturer's Certificate: Certify that products meet or exceed requirements.
  - E. Product Schedule: For panel signs. Use same designations indicated on Drawings or specified.
- 1.7 INFORMATIONAL SUBMITTALS
  - A. Qualification Data: For manufacturer.
  - B. Sample Warranty: For special warranty.



## 1.8 CLOSEOUT SUBMITTALS

- A. Maintenance Data: For signs to include in maintenance manuals.

## 1.9 MAINTENANCE MATERIAL SUBMITTALS

- A. Furnish extra signs, from the same product run, that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

## 1.10 QUALITY ASSURANCE

- A. Manufacturer Qualifications: Company specializing in manufacturing products specified in this Section with minimum five (5) years' experience.

## 1.11 WARRANTY

- A. Special Warranty: Manufacturer agrees to repair or replace components of signs that fail in materials or workmanship within specified warranty period.
  - 1. Failures include, but are not limited to, the following:
    - a. Deterioration of finishes beyond normal weathering.
    - b. Deterioration of embedded graphic image.
    - c. Separation or delamination of sheet materials and components.
  - 2. Warranty Period: Three (3) years from date of Substantial Completion.

## PART 2 - PRODUCTS

### 2.1 PERFORMANCE REQUIREMENTS

- A. Structural Performance: Signs and supporting elements are to withstand the effects of gravity and other loads within limits and under conditions indicated:
  - 1. High coastal wind gusts of 120mph at 3 seconds with minimum allowable wind pressure of 30psf.
- B. Thermal Movements: For exterior signs, allow for thermal movements from ambient and surface temperature changes.
  - 1. Temperature Change: 120 deg F (67 deg C), ambient; 180 deg F (100 deg C), material surfaces.
- C. Accessibility Standard: Comply with applicable provisions in the USDOJ's "2010 ADA Standards for Accessible Design"

## 2.2 PANEL SIGNS

- A. Panel Sign: Sign with smooth, uniform surfaces; with message and characters having uniform faces, sharp corners, and precisely formed lines and profiles.
  - 1. Solid-Sheet Sign: Aluminum sheet with finish specified in "Surface Finish and Applied Graphics" Subparagraph and as follows:
    - a. Thickness: Manufacturer's standard for size of sign.
    - b. Surface-Applied, Flat Graphics: Applied baked enamel or powder coat paint.
  - 2. Mounting: As indicated, signs will be clamped onto the fiberglass bollards, instead of drilled in anchoring, so as not to deface the fiberglass bollard. Use of U bolts with penetrations through signs at areas clear of text, to allow for reading of sign.
  - 3. Surface Finish and Applied Graphics:
    - a. Baked-Enamel or Powder-Coat Finish and Graphics: Manufacturer's standard, in color as selected by CO from manufacturer's full range.
    - b. Overcoat: Manufacturer's standard baked-on clear coating.
  - 4. Text and Typeface: as selected by CO from manufacturer's full range. Finish raised characters to contrast with background color.
  - 5. Flatness Tolerance: Sign is to remain flat or uniformly curved under installed conditions as indicated on Drawings and within a tolerance of plus or minus 1/16 inch (1.5 mm) measured diagonally from corner to corner.

## 2.3 FIELD-APPLIED, VINYL-CHARACTER SIGNS – Not Used

## 2.4 PANEL-SIGN MATERIALS

- A. Aluminum Sheet and Plate: ASTM B209 (ASTM B209M), alloy and temper recommended by aluminum producer and finisher for type of use and finish indicated.

## 2.5 ACCESSORIES

- A. Fasteners and Anchors:
  - 1. Furnish stainless steel devices as indicated in Appendix B.
    - a. Sign Mounting Fasteners:
      - 1) Stainless steel 7/8-9 in. threaded U-bolts
      - 2) Inside width/diameter of 12-7/8 in.
      - 3) Inserted into sign assembly, via drilled holes.
      - 4) Inserted into a matching length of rubber hose to prevent bolt from defacing the fiberglass bollard surfaces.
    - b. Provide stainless steel hex nuts, sized 7/8-9in

## 2.6 FABRICATION

- A. General: Provide manufacturer's standard sign assemblies according to requirements indicated.
  - 1. Preassemble signs in the shop to greatest extent possible. Disassemble signs and assemblies only as necessary for shipping and handling limitations. Clearly mark units for reassembly and installation; apply markings in locations concealed from view after final assembly.
  - 2. Internally brace signs for stability, to meet structural performance loading without oil-canning or other surface deformation, and for securing fasteners.
  - 3. Provide rabbets, lugs, and tabs necessary to assemble components and to attach to existing work. Drill and tap for required fasteners. Use concealed fasteners where possible; use exposed fasteners that match sign finish.
- B. Surface-Engraved Graphics: Machine engrave characters and other graphic devices into indicated sign surface to produce precisely formed copy, incised to uniform depth.
  - 1. Engraved Metal: Fill engraved graphics with manufacturer's standard baked enamel.

## 2.7 GENERAL FINISH REQUIREMENTS

- A. Protect mechanical finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.
- B. Appearance of Finished Work: Noticeable variations in same piece are not acceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

## 2.8 ALUMINUM FINISHES

- A. Baked-Enamel or Powder-Coat Finish: AAMA 2603 except with a minimum dry film thickness of 1.5 mils (0.04 mm). Comply with coating manufacturer's written instructions for cleaning, conversion coating, and applying and baking finish.

## PART 3 - EXECUTION

### 3.1 EXAMINATION

- A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.
- B. Verify that sign-support surfaces are within tolerances to accommodate signs without gaps or irregularities between backs of signs and support surfaces unless otherwise indicated.

- C. Verify that anchorage devices are correctly sized and located to accommodate signs.
- D. Proceed with installation only after unsatisfactory conditions have been corrected.

### 3.2 INSTALLATION

- A. General: Install signs using mounting methods indicated and according to manufacturer's written instructions.
  - 1. Install signs level, plumb, true to line, and at locations and heights indicated, with sign surfaces free of distortion and other defects in appearance.
  - 2. Install signs so they do not protrude or obstruct according to the accessibility standard.
  - 3. Before installation, verify that sign surfaces are clean and free of materials or debris that would impair installation.
  - 4. Corrosion Protection: Coat concealed surfaces of exterior aluminum in contact with grout, concrete, masonry, wood, or dissimilar metals, with a heavy coat of bituminous paint.
- B. Mounting Methods:
  - 1. Brackets: Remove loose debris from substrate surface and install U-bolts in position so that signage is correctly located and aligned.
  - 2. Each sign will be mounted with two U-bolts, one at top and other at bottom of sign, securing signs tightly to bollards.
  - 3. Do not overtighten hex nuts, so as to crack/crack fiberglass bollards.

### 3.3 ADJUSTING AND CLEANING

- A. Remove and replace damaged or deformed signs and signs that do not comply with specified requirements. Replace signs with damaged or deteriorated finishes or components that cannot be successfully repaired by finish touchup or similar minor repair procedures.
- B. Remove temporary protective coverings and strippable films as signs are installed.
- C. On completion of installation, clean exposed surfaces of signs according to manufacturer's written instructions, and touch up minor nicks and abrasions in finish. Maintain signs in a clean condition during construction and protect from damage until acceptance by Owner.

**END OF SECTION 10 14 23**

## **SECTION 31 62 23 – COMPOSITE PILES**

### **PART 1 - GENERAL**

#### **1.1 SUMMARY**

- A. Section includes driven fiberglass piling.

#### **1.2 UNIT PRICES - MEASUREMENT AND PAYMENT**

- A. Actual Piles:

1. Actual Pile Quantity: 23
2. Actual Pile Length: 15 linear feet
3. Actual Pile Diameter: 10 - 12 inches
4. Actual Pile Bury Depth: 8 linear feet

#### **1.3 PERFORMANCE REQUIREMENTS**

- A. Drive piles to be vertically plumb, in line with existing bollards, and in a uniformed consistent spacing as to provide for a pleasing consistent aesthetic view shed.

#### **1.4 SUBMITTALS**

- A. Section 013300 - Submittal Procedures: Requirements for submittals.
- B. Shop Drawings: Indicate details and schedules of pile installation, identify recommended pile lengths and diameters to suit design requirements.
- C. Product Data:
  1. Submit concrete mix design indicating cured properties and mix components.
  2. Submit data for fiberglass sleeves, shells, reinforcement, and pile cushion caps.
- D. Manufacturer's Certificate: Certify products meet or exceed specified requirements.

#### **1.5 CLOSEOUT SUBMITTALS**

- A. Section 017000 - Execution and Closeout Requirements: Requirements for submittals.
- B. Project Record Documents: Accurately record the following:
  1. Sizes, lengths, and locations of piles.
  2. Sequence of driving.
  3. Drilling: Hole diameters.
  4. Final base and head elevations.

## 1.6 QUALITY ASSURANCE

- A. Perform Work in accordance with State of Texas standard.

## 1.7 QUALIFICATIONS

- A. Installer: Company specializing in performing Work of this section with minimum 5 years documented experience.
- B. Design and select pile components under direct supervision of Professional Engineer experienced in design of this Work and licensed in State of Texas.
- C. Monitor pile driving operations by Geotechnical Engineer experienced in this Work and licensed in State of Texas.

## 1.8 PRE-INSTALLATION MEETINGS

- A. Section 013000 - Administrative Requirements: Pre-installation meeting.
- B. Convene minimum one week prior to commencing Work of this section.

# PART 2 - PRODUCTS

## 2.1 MATERIALS

- A. Resin Matrix Fiberglass Shells: An epoxy/vinylester, with high elongation, low styrene monomer, excellent hydrolytic stability and high heat deflection temperature, will not leach, inert, high chemical resistance, insoluble in any common hydrocarbons, mild acidic or alkaline solutions.
- B. Reinforcement: A quasi-isotropic, three-dimensional proprietary fabric utilizing e-glass grade filaments yielding exceptional damage tolerance without crack propagation.
- C. Outer Layer: Coating is a dual component, polyurethane paint system. This technology provides for excellent chemical and weather resistance along with superior durability.
- D. Pile Cushion Cap: Shall be chemical and weather resistant. End caps shall protect bollards from internal rot and splintering and have a 10-year warranty. End caps shall be secured to the bollards by using screws or as recommended by the manufacturer.
  - 1. Color: Black to match existing
  - 2. Shape: Cone
- E. Refer to Appendix C for material tolerances and other specifications.
  - 1. Pearson Pilings or approved equal.

## PART 3 - EXECUTION

### 3.1 EXAMINATION

- A. Section 013000 - Administrative Requirements: Verification of existing conditions before starting work.

### 3.2 PREPARATION

- A. Use installation method which will not cause damage to nearby structures.
- B. Protect structures near the Work from damage.

### 3.3 INSTALLATION

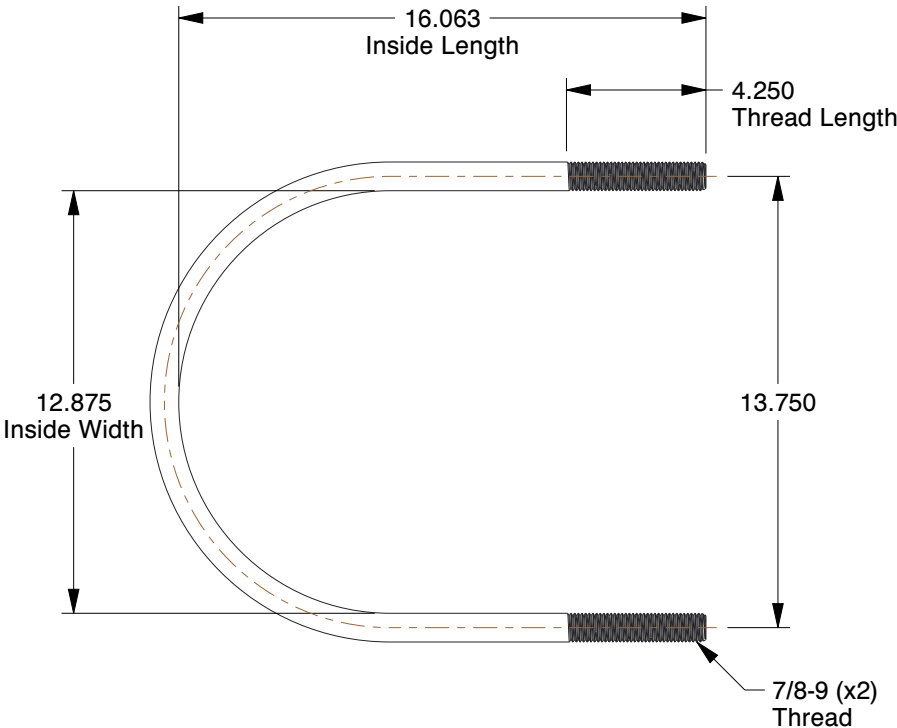
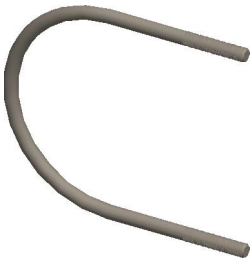
- A. Install per manufacturer instructions.
- B. Hydro-jet utilizing adjacent salt-water sources to reduce impacts to surrounding natural resources at two separate locations.
- C. Do not damage piles during driving operations.
- D. Set tops of piles to elevations indicated.
- E. Allow inspection of pile prior to placement of next item.
- F. Prepare pile top to receive pile cap.

### 3.4 FIELD QUALITY CONTROL

- A. Section 01 40 00 Quality Requirements
- B. 01 70 00 Execution and Closeout Requirements: Field inspecting, testing, adjusting, and balancing.
- C. Test Piles: Same diameter and type as specified for other piling, placed in same manner.
- D. Accepted test piles may be used in the Work.
- E. Unacceptable Piles: Piles that fail tests, are placed out of position, are below cut-off elevations, or are damaged.
- F. Provide additional piles or replace piles to conform to specified requirements.

**END OF SECTION 31 62 23**

Appendix A- U bolt Grainger



NOTES:

- 1. MATERIAL : 304 Stainless Steel
- 2. FINISH : Plain
- 3. TYPE : Long Tangent
- 4. INCLUDES : 4 Hex Nuts

| COMPONENT                                                     |  | U-Bolt |  |
|---------------------------------------------------------------|--|--------|--|
| <div>5ZB20</div> <div>U Bolt, 304 SS, 7/8-9, Pipe 12 In</div> |  | UNIT   |  |
|                                                               |  | INCH   |  |
|                                                               |  | SHEET  |  |
|                                                               |  | 1 of 1 |  |

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SCALE 0.17



# Appendix B Pearson Pilings reference sheet

## MATERIAL PROPERTIES

### Reinforcement

A quasi-isotropic, three-dimensional proprietary fabric utilizing e-glass grade filaments yielding exceptional damage tolerance without crack propagation.

### Resin Matrix

An epoxy/vinylester, with high elongation, low styrene monomer, excellent hydrolytic stability and high heat deflection temperature, will not leach, inert, high chemical resistance, insoluble in any common hydrocarbons, mild acidic or alkaline solutions.

### Outer Layer

Coating is a dual component, polyurethane paint system. This technology provides for excellent chemical and weather resistance along with superior durability.

## MECHANICAL PROPERTIES

| MINIMUM VALUES*                             | PILING DIAMETER |           |           |           |           |           |
|---------------------------------------------|-----------------|-----------|-----------|-----------|-----------|-----------|
|                                             | 8"              | 10"       | 12"       | 12" HD    | 14"       | 14" HD    |
| Axial Tensile Strength – psi                | 50,123          | 55,622    | 61,565    | 68,000    | 66,890    | 66,650    |
| Axial Tensile Modulus (MOE) – psi           | 4,547,780       | 3,827,560 | 4,587,990 | 4,370,000 | 4,320,000 | 4,555,000 |
| Axial Flexural Strength – psi               | 58,400          | 58,636    | 68,136    | 89,400    | 75,600    | 81,530    |
| Axial Compressive Strength – psi            | 43,320          | 58,919    | 54,289    | 66,546    | 66,402    | 75,240    |
| Transverse Tensile Strength – psi           | 11,733          | 21,604    | 24,271    | 23,605    | 17,411    | 22,600    |
| Transverse Tensile Modulus – psi            | 2,433,870       | 2,276,160 | 2,699,950 | 3,360,000 | 6,727,000 | 6,840,000 |
| Interlaminar Shear Strength – psi           | 13,000          | 12,000    | 12,000    | 12,000    | 12,000    | 12,000    |
| Effective Bending Stiffness – psi           | 1.62E+8         | 3.72E+8   | 1.06E+9   | 1.31E+9   | 1.61E+9   | 2.85E+9   |
| Young's Modulus                             | 4,547,780       | 3,827,560 | 4,587,990 | 4,370,000 | 4,320,000 | 4,555,000 |
| Poisson's Ratio                             | 0.22            | 0.25      | 0.23      | 0.23      | 0.24      | 0.24      |
| Allowable Bending Moment – kips-ft (FS = 2) | 11              | 31        | 78        | 144       | 94        | 218       |
| Barcol Hardness                             | >50             | >50       | >50       | >50       | >50       | >50       |
| Glass to Resin Ratio – by weight            | ~60:40          | ~60:40    | ~60:40    | ~60:40    | ~60:40    | ~60:40    |
| Aprox. Wall Thickness – in                  | ~.117           | ~0.250    | ~0.375    | ~0.500    | ~0.375    | ~0.50     |
| Aprox. Weight – lbs/ft                      | 4               | 7         | 10        | 16        | 13        | 17        |
| Thermal Expansion – in/in/°F                | <.000014        | <.000012  | <.000006  | <.000006  | <.000006  | <.000006  |
| Water Absorption – %                        | <.25            | <.25      | <.25      | <.25      | <.25      | <.25      |
| Reinforcement – (g/m2)                      | 3496            | 5312      | 8300      | 11230     | 9346      | 13400     |

\*Pearson Pilings does have the ability to customize laminates to meet the needs of your project should requirements exceed our standard piling specifications\*